

ANALOG & DIGITAL CIRCUITS

Lecture 10

Computer Science Department, TUCN

-

CONVERTERS

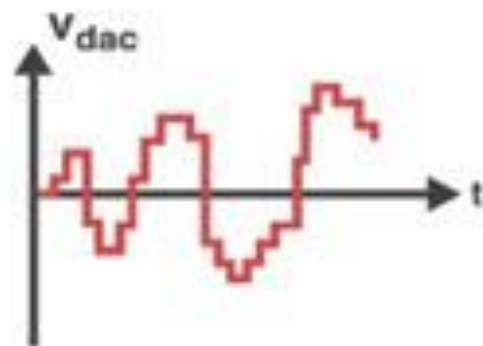
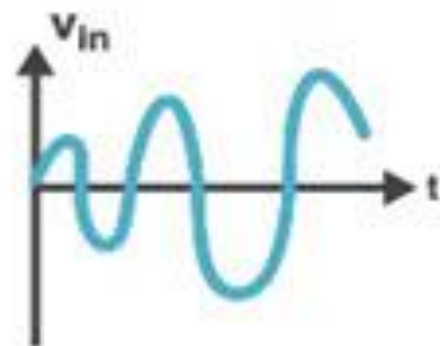
Sampling

Signal digitization

Analog-to-Digital Converters

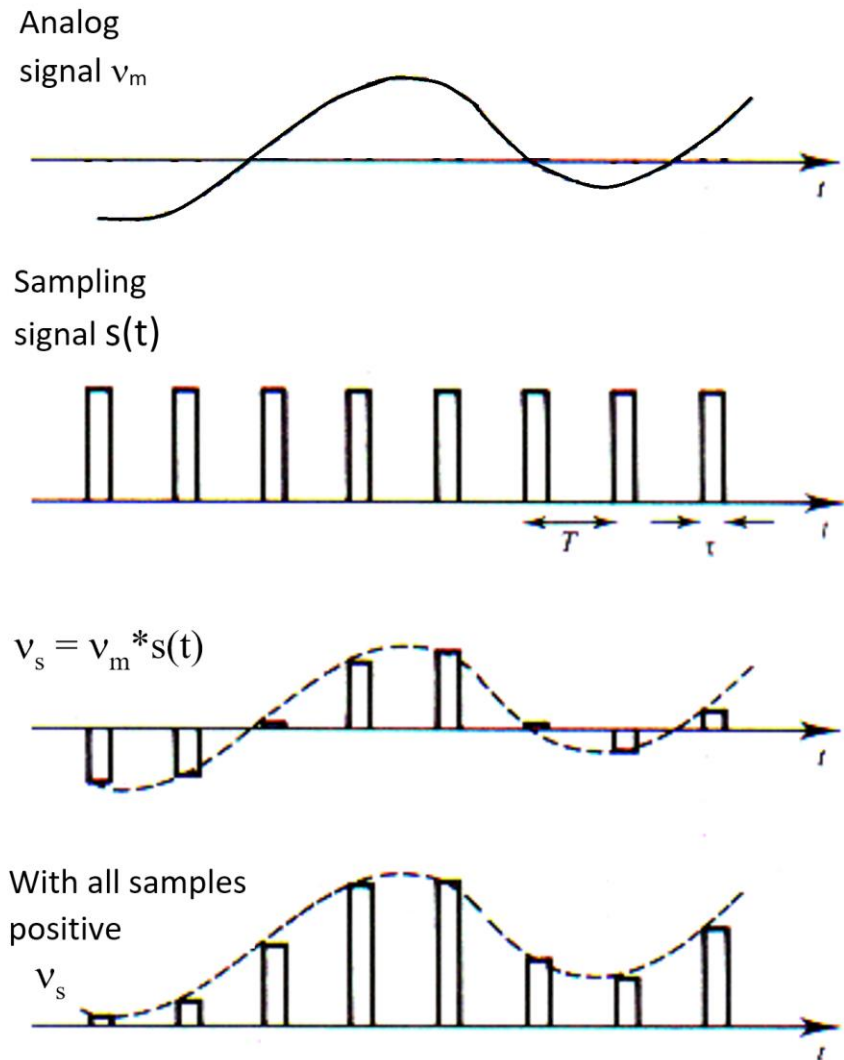
Digital-to-Analog Converters

Need of conversion



Sampling and quantization

- Basic process of **sampling** is the gating of an analogue signal by a periodic pulse which will only allow the signal through whilst each pulse is on.
- The gating signal or the sampling function $s(t)$ has pulses of constant height, length (τ) and separation time (T).
- Ideal, τ it should be as small as possible and the sampling frequency as high as possible



- The analog baseband signal $v_m(t)$ is usually level shifted, so no part of it is negative; therefore, all samples will be positive
- Multiplying the baseband signal $v_m(t)$ by the sampling signal $s(t)$, the sampled signal v_s is obtained, made from slices (samples). $v_s = v_m * s(t)$
- T (T_s) is the sampling interval, $f_s = 1/T_s$ is the sampling frequency and τ is the sampling time.

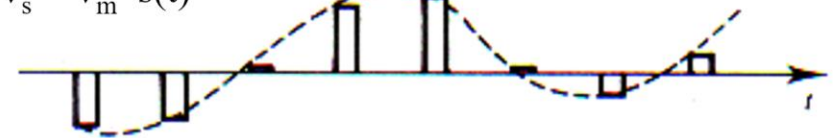
Analog
signal v_m



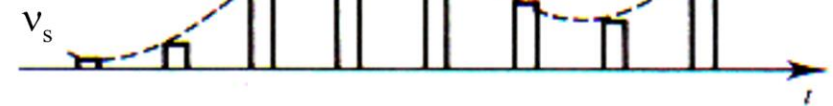
Sampling
signal $s(t)$



$v_s = v_m * s(t)$



With all samples
positive

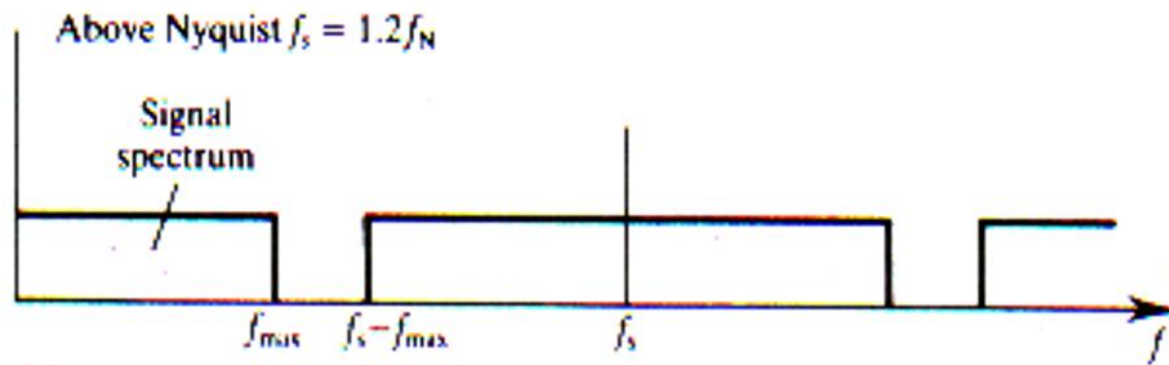


- The result of sampling is the function:

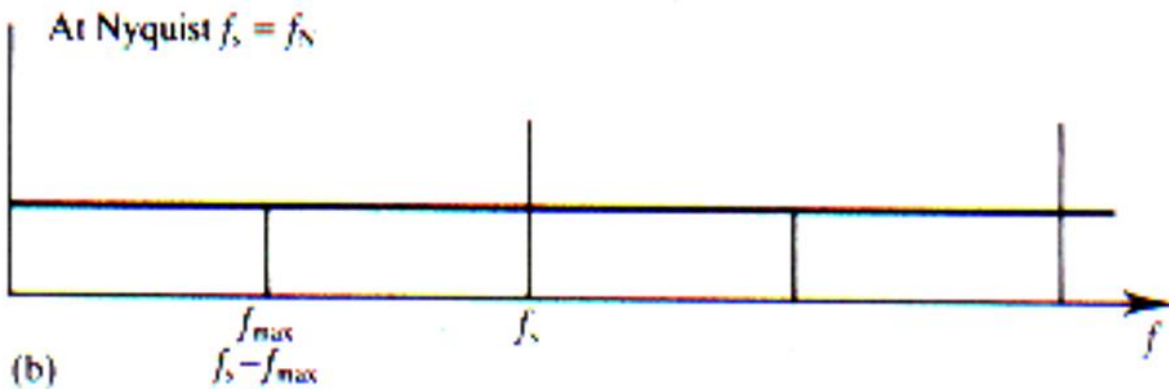
$$v_s = v_m s(t) = (A_m \tau / T) [1 + \cos \omega_m t + \cos(\omega_s - \omega_m)t + 2\cos \omega_s t + \cos(\omega_s + \omega_m)t + \cos(2\omega_s - \omega_m)t + 2\cos 2\omega_s t + \cos(\omega_s + \omega_m)t + \dots]$$

- The spectrum is given by the next slide.
- To recover the original baseband by filtering, the **Nyquist criterion** must be met.
- The **sampling rate** must be **at least twice** the highest baseband frequency:

$$f_s \geq 2 f_m (= f_N), \text{ where } f_N \text{ is said to be Nyquist frequency.}$$
- Practically, if the sampling occurs at too slow a rate, the lower sideband ($f_s - f_m$) of the sampling frequency overlaps the baseband, corrupting it. Nyquist criterion is illustrated by the figure next slides.



(a)



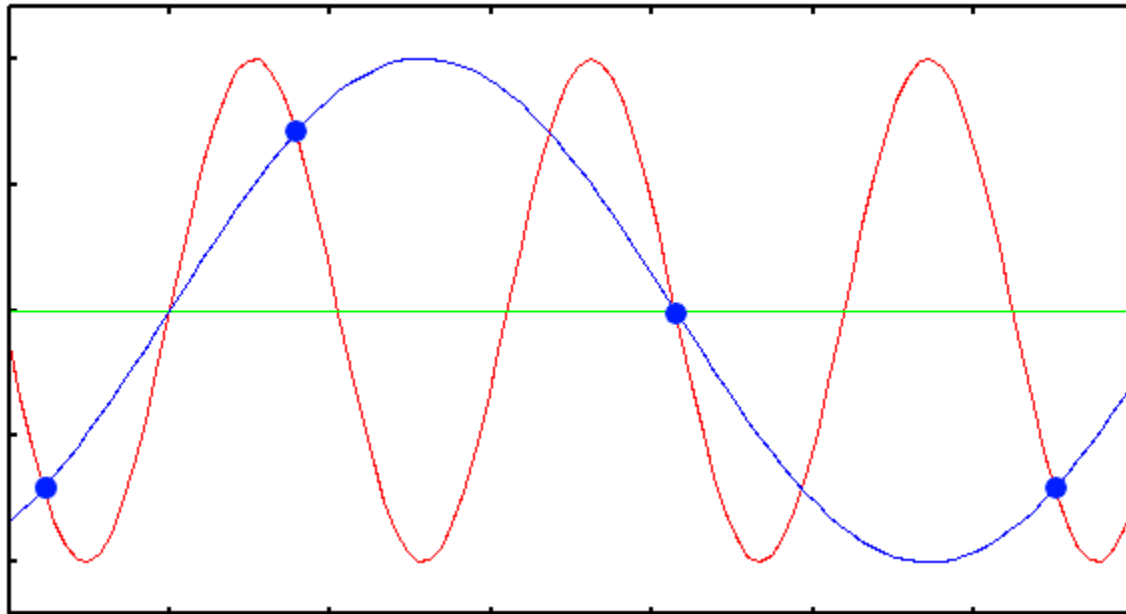
(b)



(c)

Nyquist criterion illustration

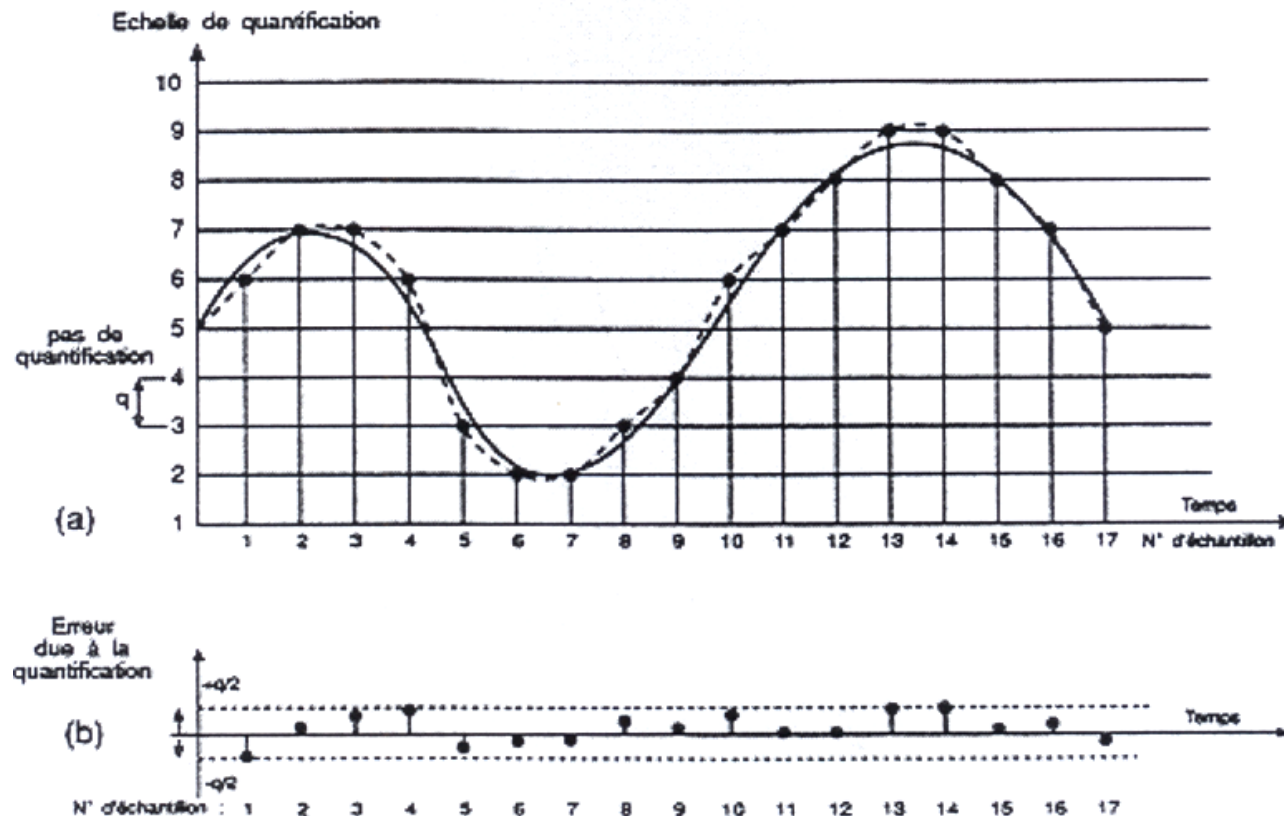
- Another problem avoided using this criterion is **aliasing**; a lower sideband of f_s can appear within the baseband range and be thought to be part of it (disguised itself, taking a false name – alias).
- It happens when the original signal is affected by noise (presents higher frequency than highest signal frequency); use of low-pass filter or anti-aliasing filter.



Plot showing aliasing of an undersampled 1D sinusoidal signal. The blue dots are the digital samples taken to record the red signal. Clearly, they are not enough to reconstruct the original signal: from then on, only the blue, wrong signal can be interpolated. (<https://svi.nl/AliasingArtifacts>)

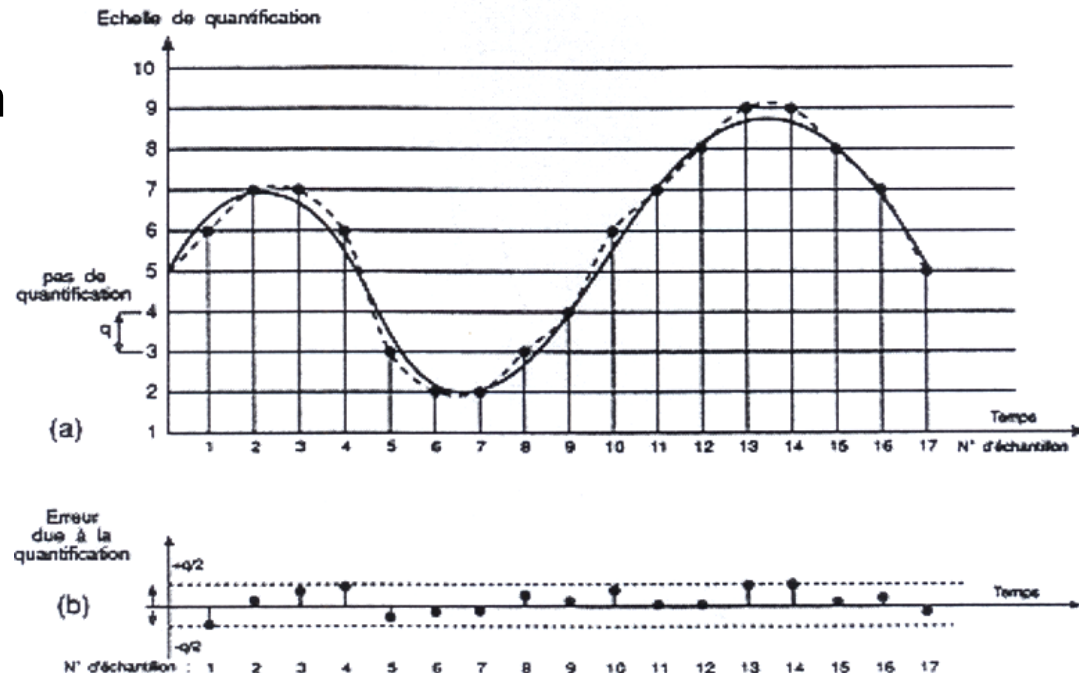
Signal digitization (Quantization)

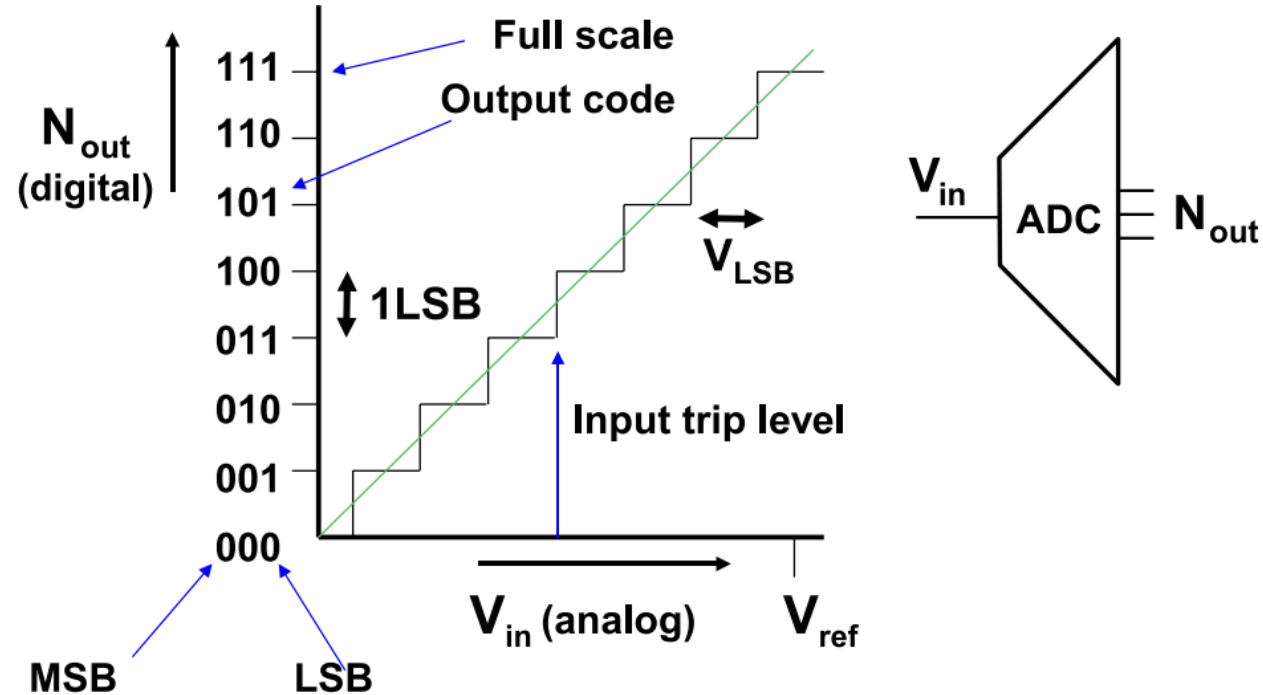
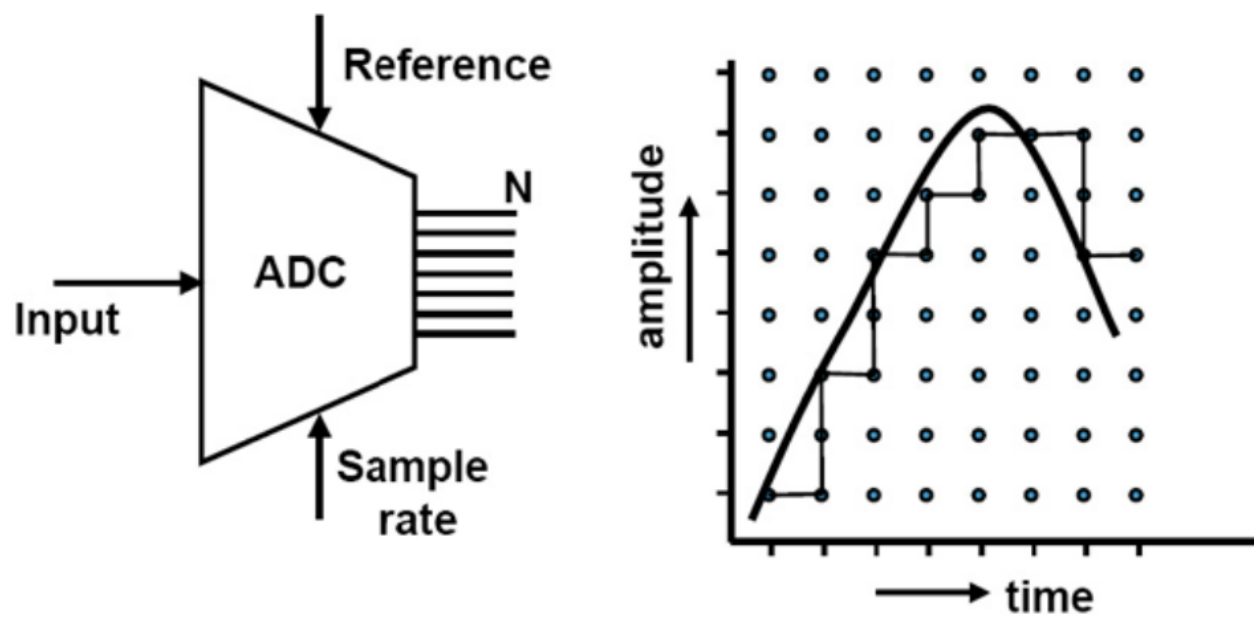
- *Quantization is an interpretation of a continuous quantity by a **finite set of discrete values**; means establishing numerical (binary) values, starting from an analog signal value.*
- Using N bits, may obtain 2^N levels; each value of each sample will have associated a N bit binary value; **the higher N is, the better the accuracy.**
- The power N is called the resolution of a conversion, and defines the number of levels to which an amplitude continuous signal can be rounded as 2^N

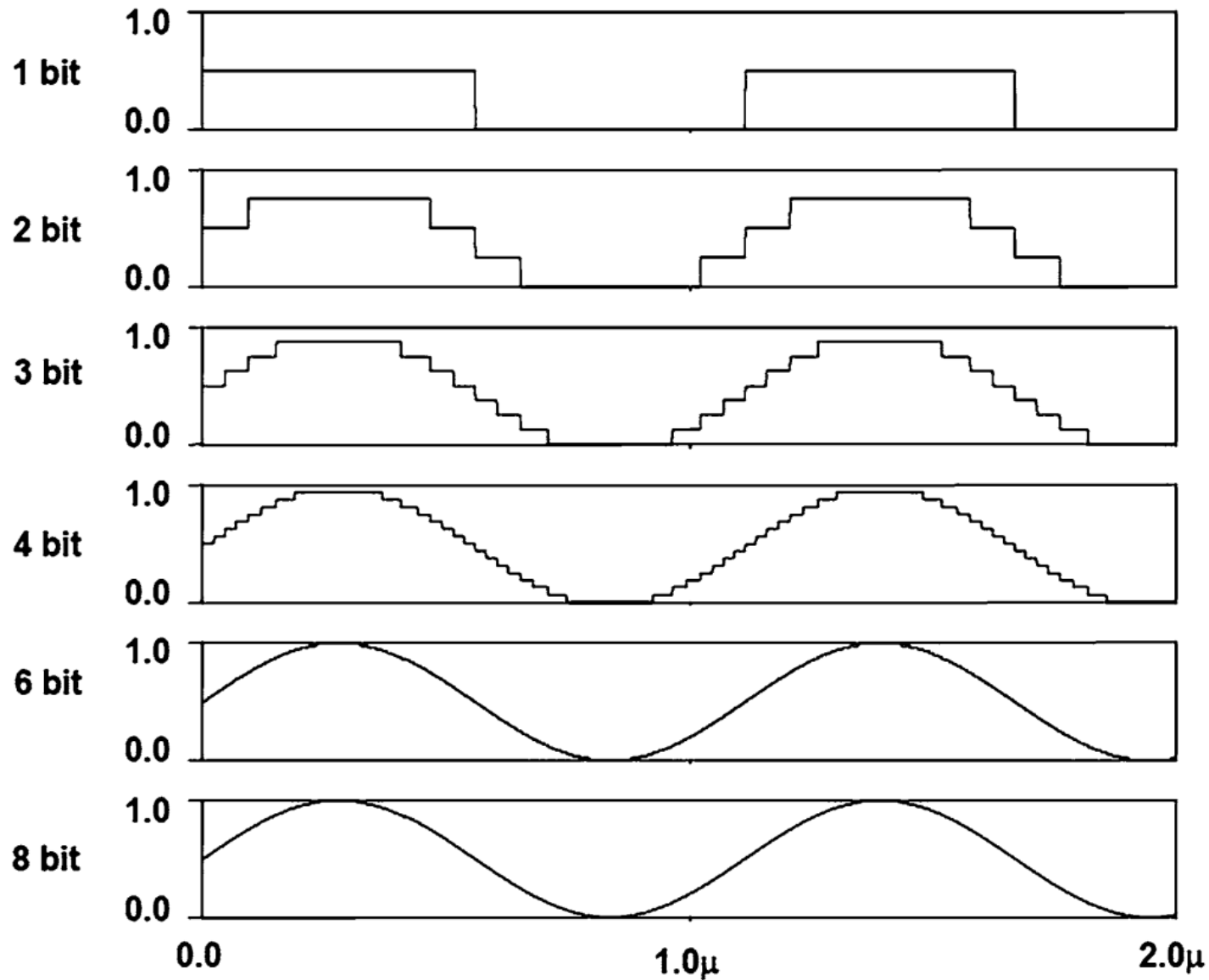


Signal digitization (Quantization)

- Amplitude quantization approximates its input by a discrete amplitude taken from finite set of values
- **Quantization step size** will be done by:
$$Q_s = \frac{V_{\max} - V_{\min}}{2^N}$$
- **Quantization error** means the difference between the signal's value and the associated binary value.
- The error can be given both by the sampling and retention process (the value of the current sample is maintained as a signal value until the next sampling) and by the quantization process, which raises or lowers the sampled value, as dictated by the digital representation



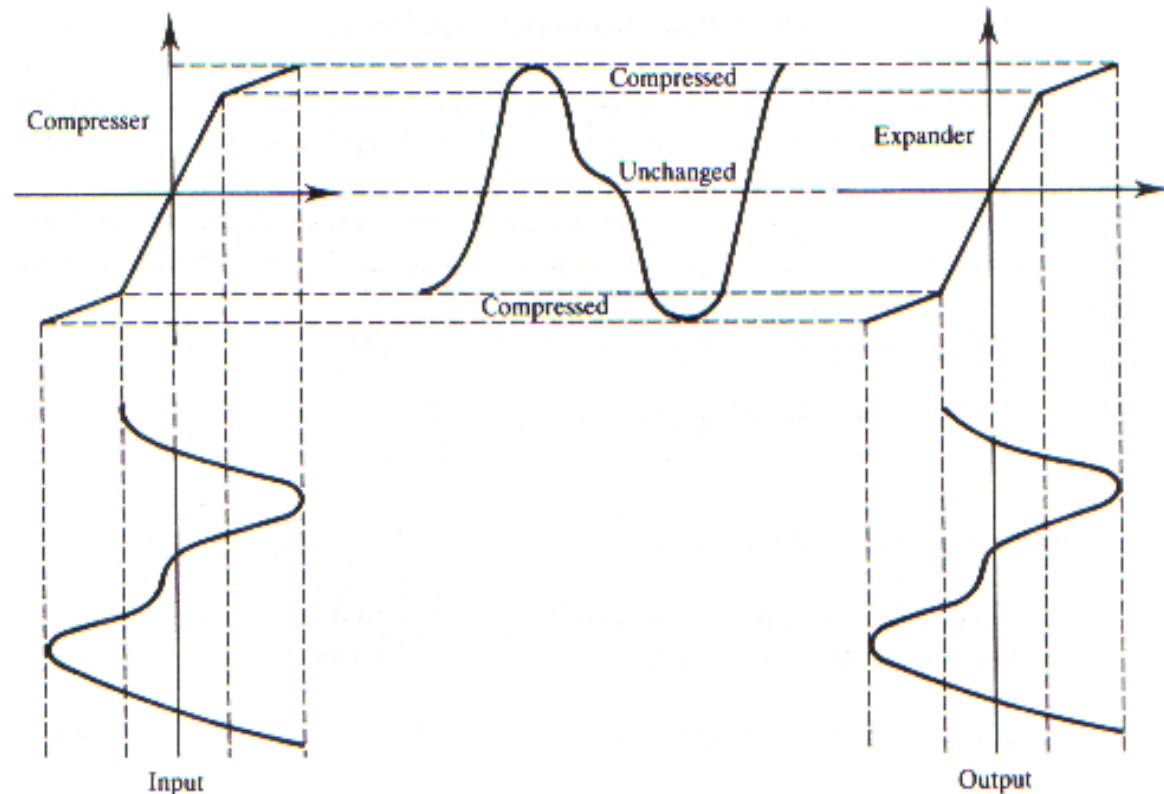




Representation of a sine wave (900 kHz) quantized with 1, 2, 3, 4, 6, and 8 bit resolution

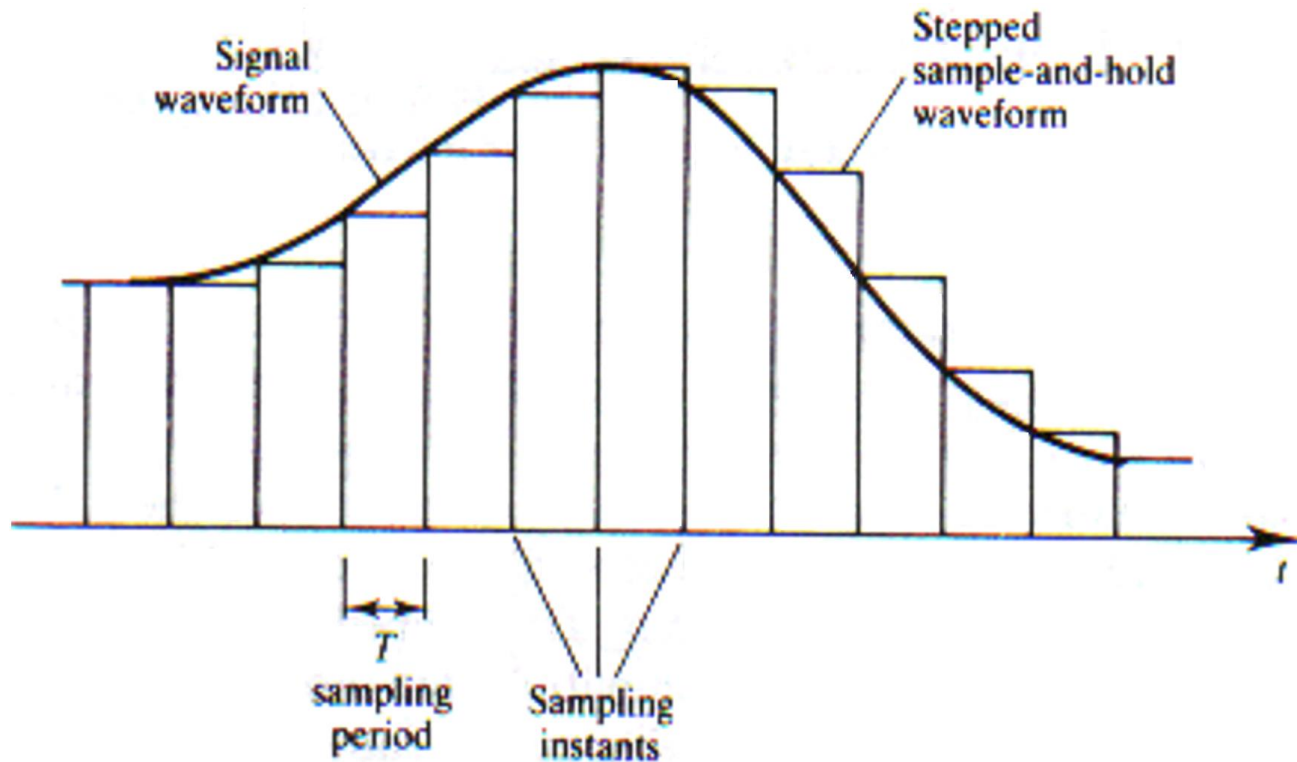
Source: Marcel J.M. Pelgrom, "Analog-to-Digital Conversion", Springer Netherlands, 2010.

- Smaller amplitude signals are subject to more quantization errors
- One solution is to increase the number of bits allocated to the representation of a sample, but this number is usually 8 or 12, rarely higher.
- The adaptive solution is combining compression and expansion techniques, a technique called companding.
- Prior to quantization, the signal is distorted, meaning that its high-value part is compressed according to a certain law, and at the receiver, when the original signal is restored, these intervals are expanded accordingly to the same laws.

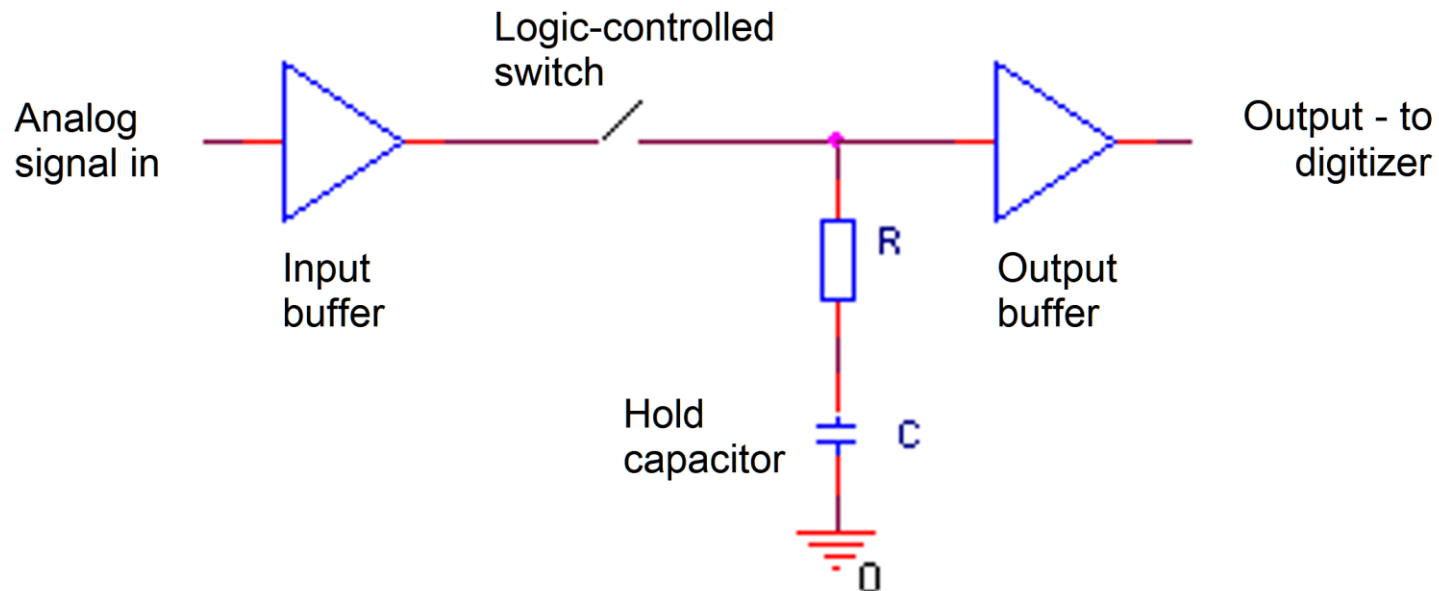


Sample and Hold

- The sampling and hold (S/H) process is defined as maintaining the value of the sample amplitude until the next sampling occurs.
- The result is a stepped waveform as in figure.



- An example of S/H circuit is given below:
- The role of the capacitor is to be charged quickly (during the sampling time) and then to hold the sampled voltage until the next sample has to replace it
- Input buffer (amplifier) offers a high input impedance to the analog signal and a low output impedance for a fast charge of the capacitor C
- The output buffer has a high input impedance, denying the hold capacitor to discharge, so having a constant value at its input



CONVERTERS

Sampling

Signal digitization

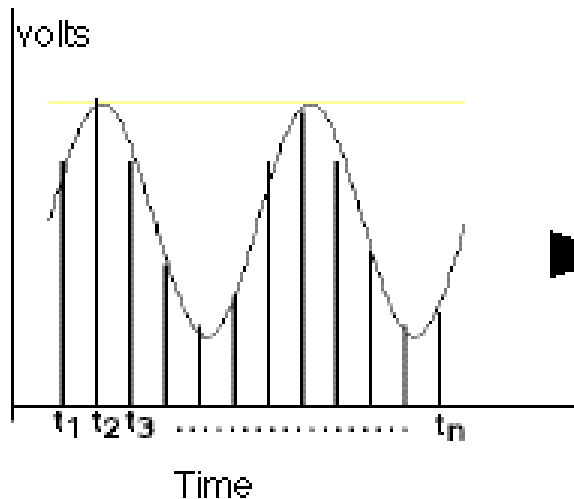
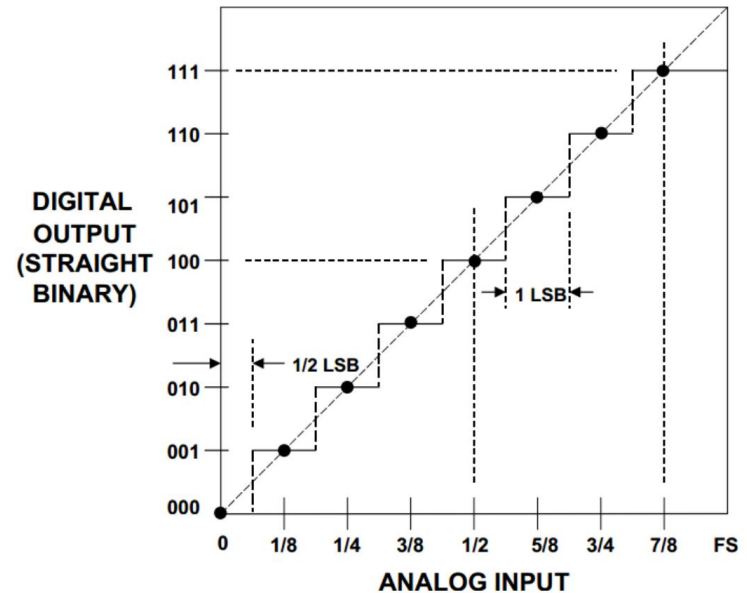
Analog-to-Digital Converters

Digital-to-Analog Converters

Analog-to-Digital Converters

- Analog to Digital Converter is transforming an analog input signal into a digital output signal one, quantified using a number of bits, at regular sampling periods
- Any ADC contains an analog input and a digital (binary) output

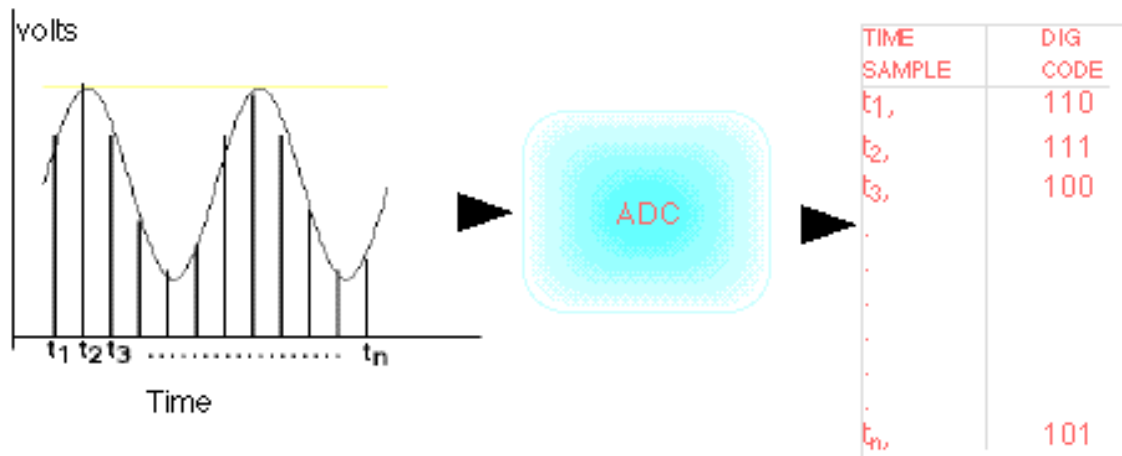
3-bit ADC example



TIME SAMPLE	DIG CODE
$t_1,$	110
$t_2,$	111
$t_3,$	100
$\vdots$	
$t_n,$	101

Analog-to-Digital Converters

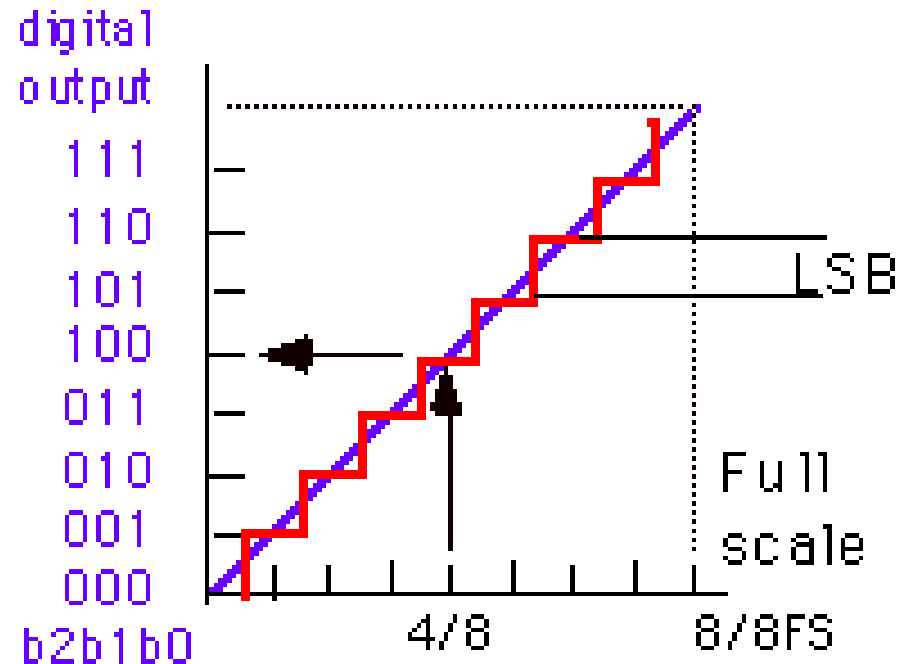
- ADC features:
 - AD conversion takes longer than DA conversion
 - An ADC may contain a DAC
 - Conversion is timed by a clock signal
 - ADC control unit is responsible for conversion, signal generation and data storage
 - ADC *resolution* (quantum) given by the lowest input variation giving a code (output) change (dependent upon the number of bits the ADC uses)
 - ADC presents an inherent quantification error, but also: offset, gain, linearity (integral and differential), hysteresis and monotony errors



- The transfer characteristic is stepped shaped (see it for a 3bit ADC)
- LSB or least significant bit is defined as the minimum increment of the voltage that a ADC can convert; LSB varies with the operating input voltage range of the ADC
- FS stands for **full scale**; represents the range of input variation
- If FS of the input signal is 10V then the LSB for a 3-bit ADC corresponds to $10/2^3=1.25V$. That is not very good!
- However, for a 12-bit ADC the least significant bit will be $10/2^{12} = 10/4096 = 2.44mV$

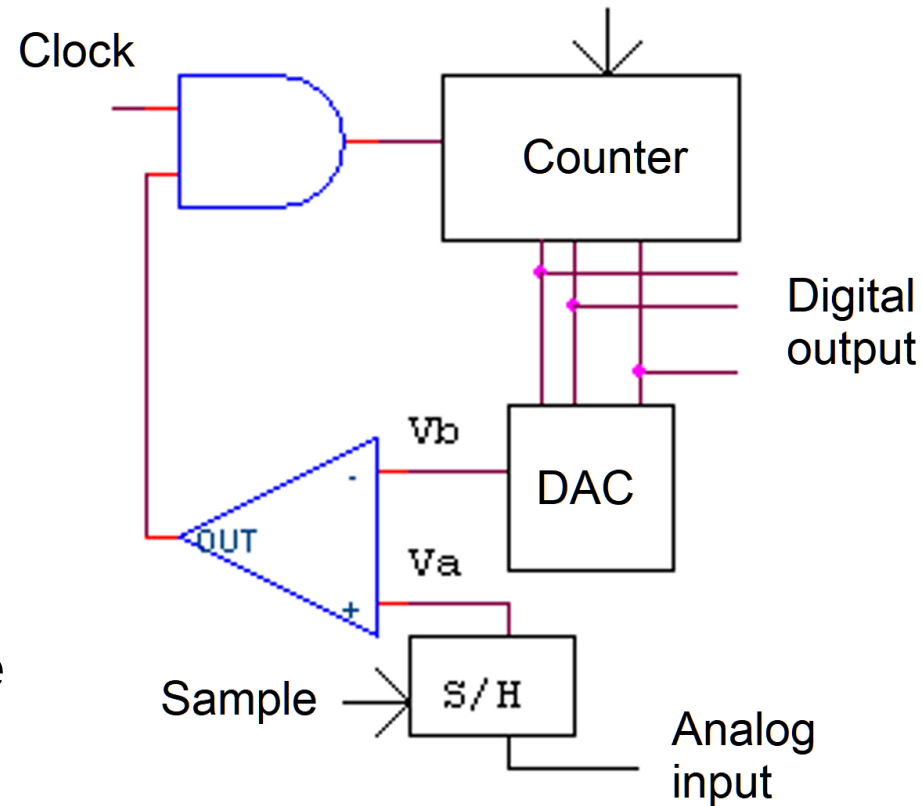
- Main ADC types are:

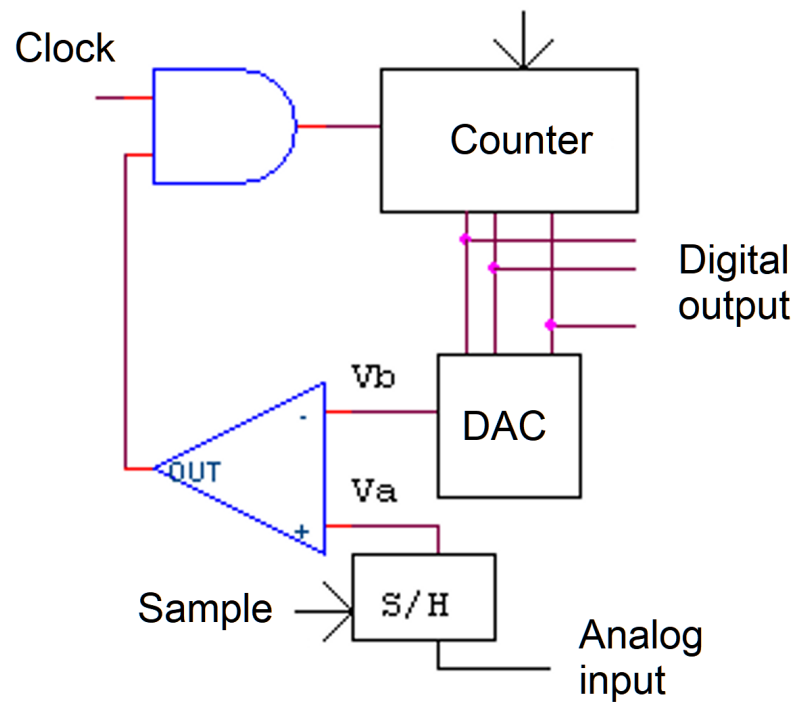
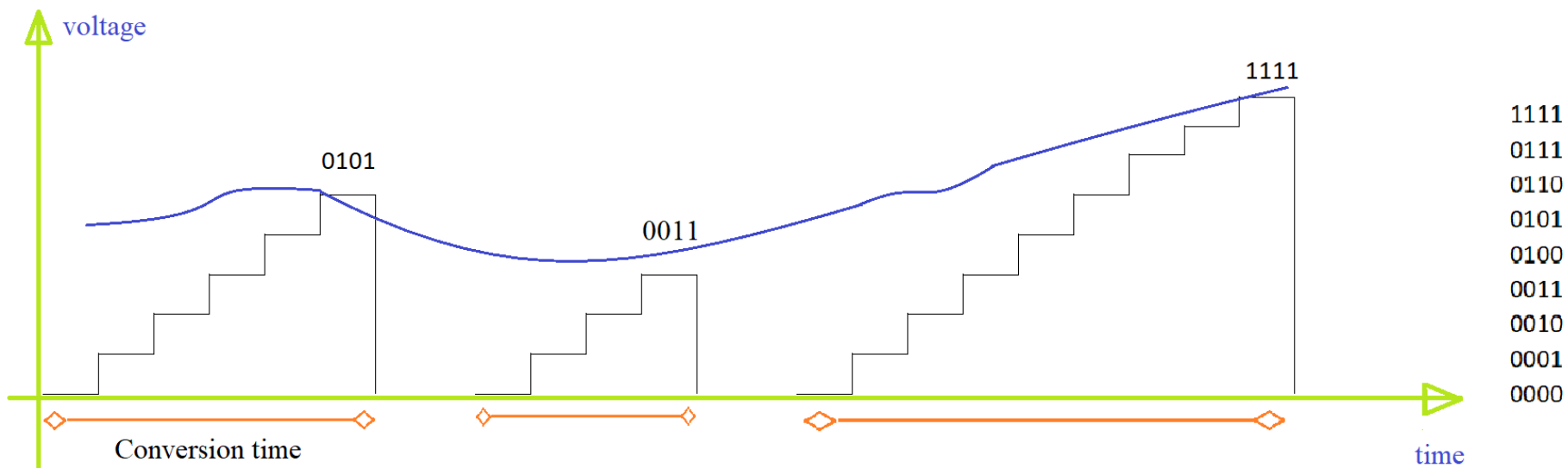
- Single slope or ramp
- Successive approximation
- Dual slope
- Parallel or 'flash'
- Sigma Delta



Single slope (ramp) ADC

- Composed of three basic elements:
 - A binary counter
 - A digital-to-analog converter
 - An analog comparator
- Operation
 - Counter is reset
 - Analog input is sampled
 - While $V_a > V_b$ the counter is incremented
 - When $V_a = V_b$ the counter stops and the binary code is available at the output
- Relatively simple**, but **slow**, since conversion time could be **up to $2N$** , where N is the resolution of the ADC

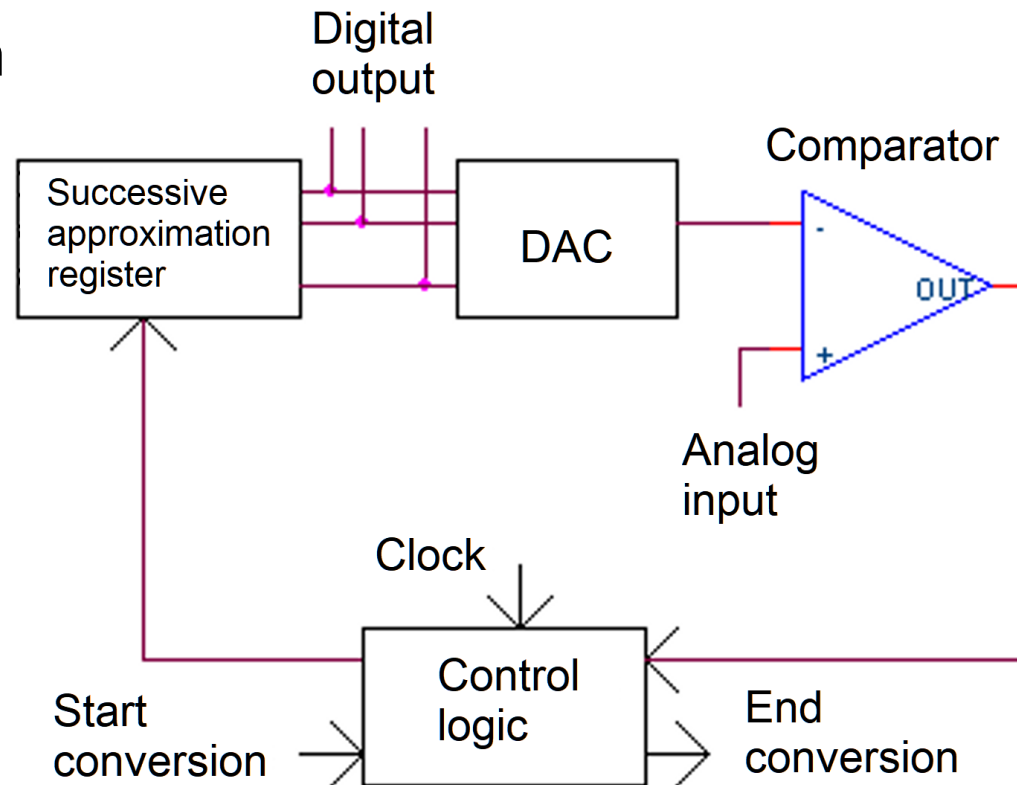




Successive approximation ADC

- The most common, based on 8 or 12 bits and contains the following basic elements:

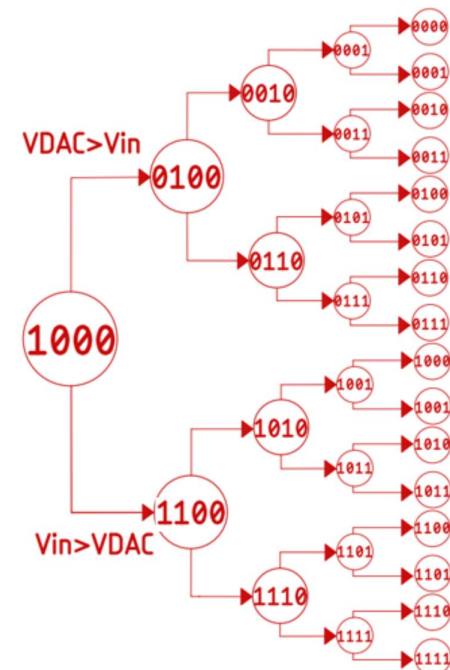
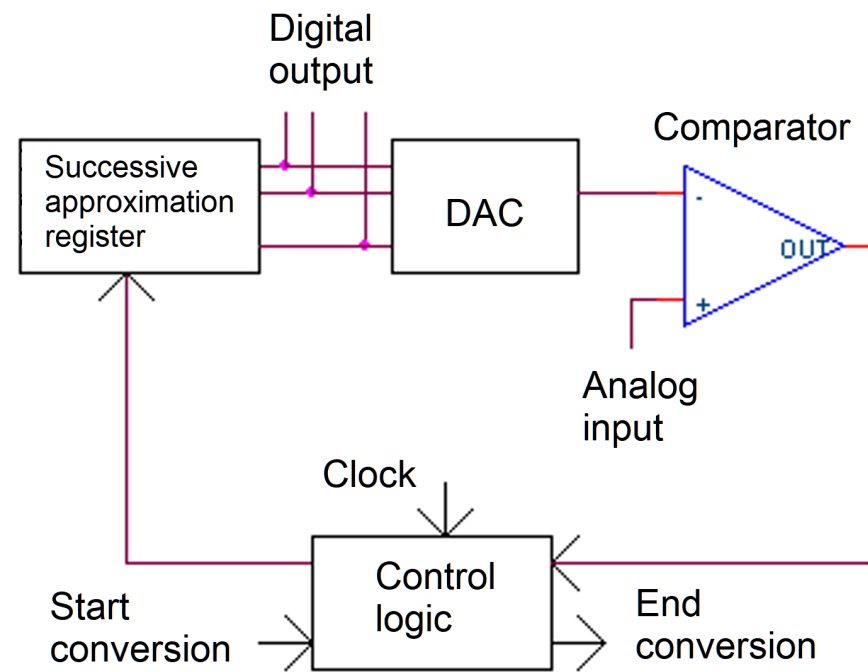
- A digital-to-analog converter (DAC)
- An analog comparator
- A control logic module
- A successive approx. register

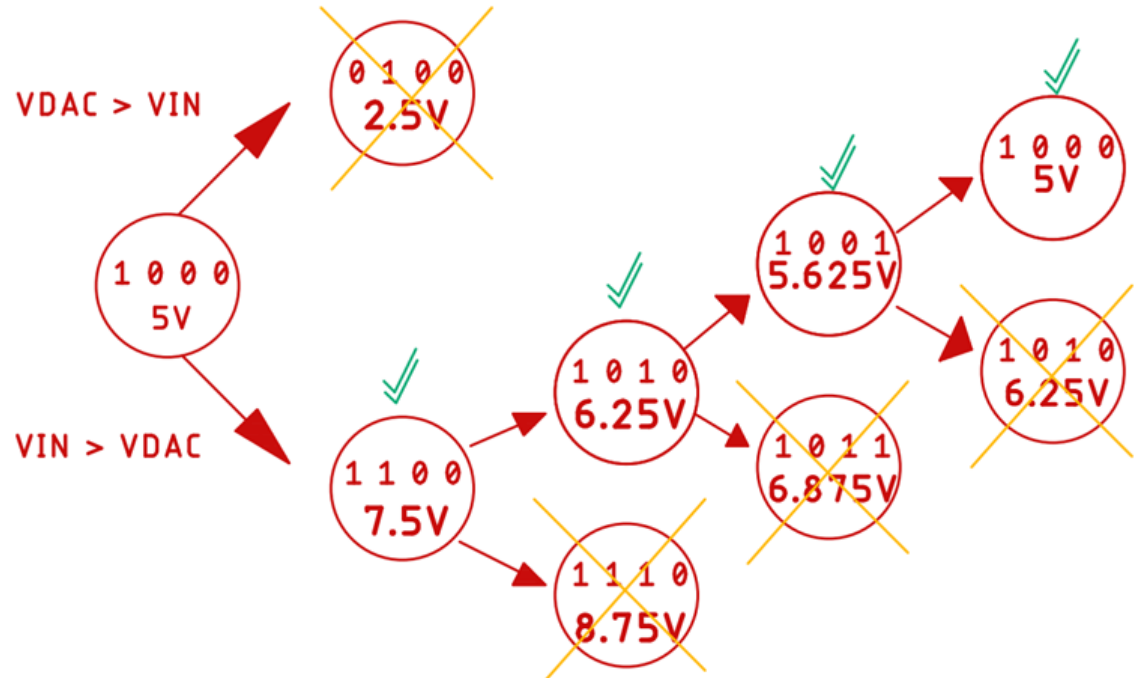
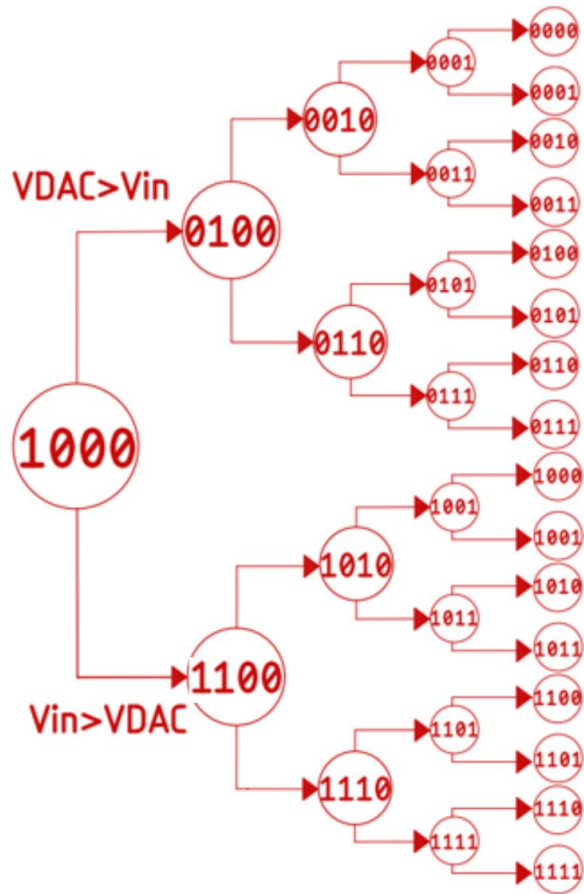


- **Characteristics**

- Conversion requires only N steps, where N is the resolution of the ADC
- Conversion times of μs are typical
- Not expansive, relatively accurate and fast

- Operation is based on a binary search:
 - Initially, the **register** provides an **output** corresponding to **half the range**, by setting the most significant bit to 1 and all others to 0 (1000...0)
 - If the analog input present at the comparator terminal is greater than the one obtained from the output of the digital-to-analog converter, MSB = 1, else MSB = 0
 - The register performs the same operation for each bit, from MSB to LSB; at the end, the register shows the numerical value of the conversion output.



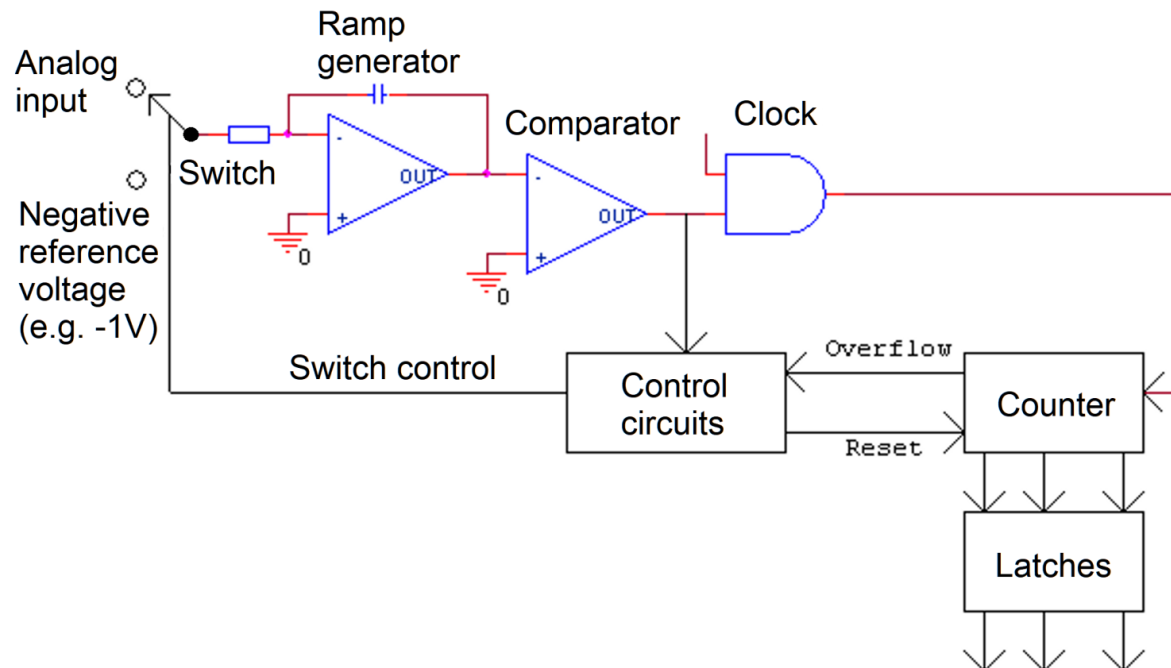


Example: 4-bit ADC, sampled input signal: 5.3V
 - reference of the ADC: 10V

Decimal	Binary	Voltage levels
15	1111	[9.375 , 10]
14	1110	[8.75 , 9.375]
13	1101	[8.125 , 8.75]
12	1100	[7.5 , 8.125]
11	1011	[6.875 , 7.5]
10	1010	[6.25 , 6.875]
9	1001	[5.625 , 6.25]
8	1000	[5 , 5.625]
7	0111	[4.375 , 5]
6	0110	[3.75 , 4.375]
5	0101	[3.125 , 3.75]
4	0100	[2.5 , 3.125]
3	0011	[1.875 , 2.5]
2	0010	[1.25 , 1.875]
1	0001	[0.625 , 1.25]
0	0000	[0 , 0.625]

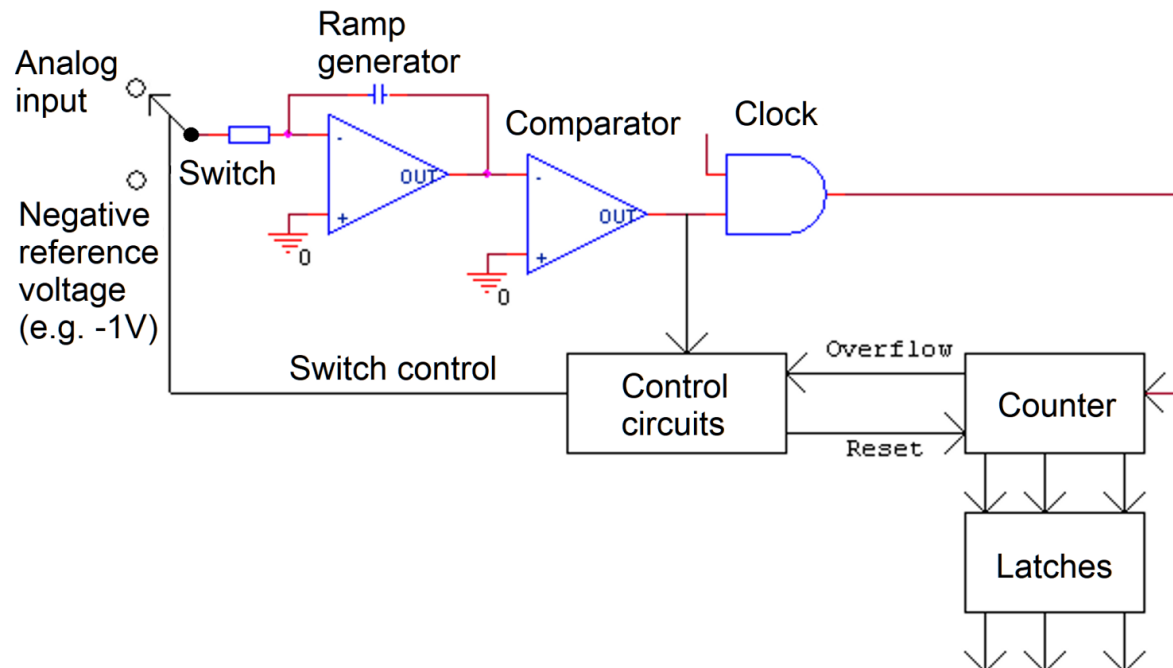
Dual slope ADC

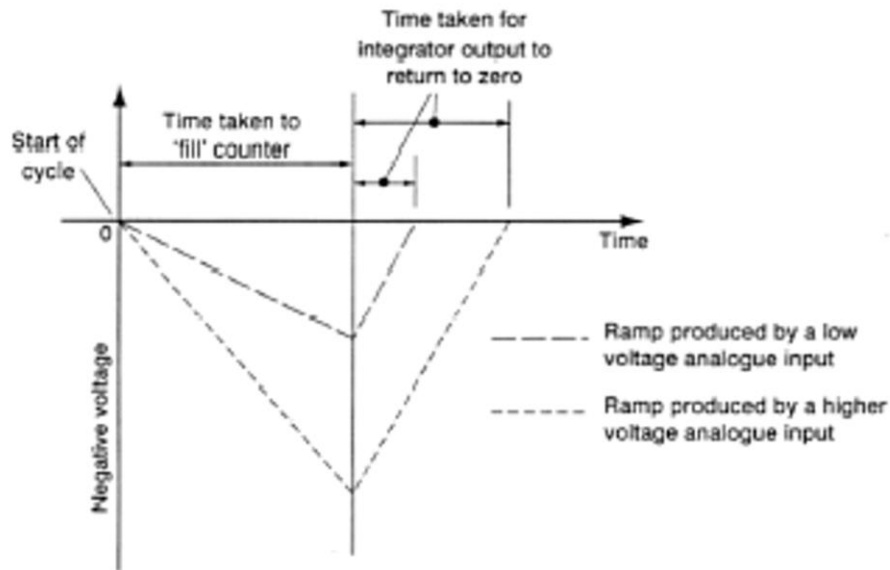
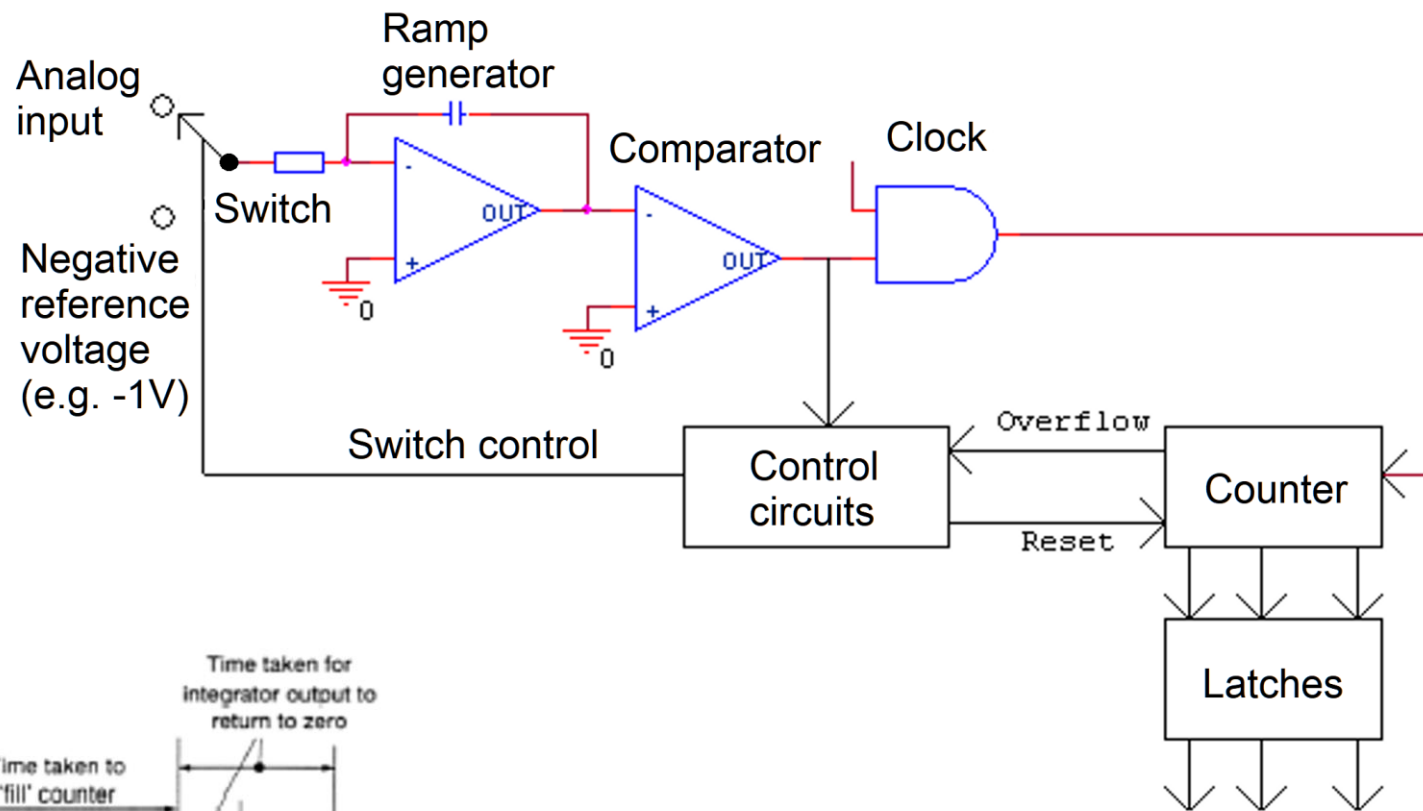
- Basic elements
 - An integrator
 - A zero-crossing detector
 - A binary counter
 - Logic gates and switches
- Characteristics
 - **Very high resolution**, but also **slower** (30 conversions/sec)
 - Widely used in digital multimeters
 - Insensitive to clock drift, RC drifts and high-frequency noise



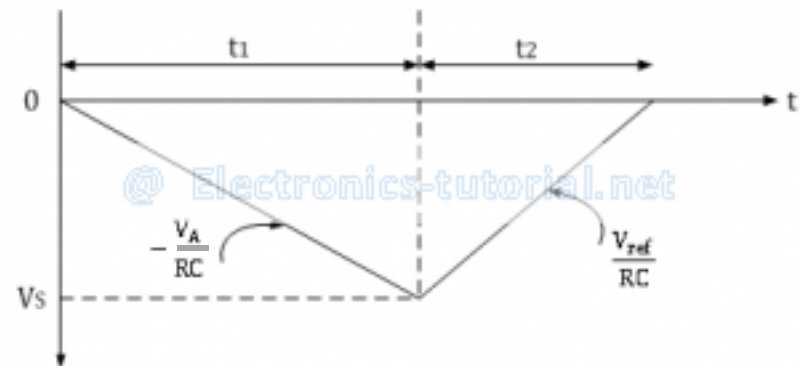
- Operation

- Counter is reset and the switch is connected to the analog input
- The integrator generates a negative ramp whose slope is proportional to the analog input
- The comparator goes HIGH, enabling clock pulses into the counter
- When counter overflows, it resets to zero and the control circuit switches the switch to a reference negative voltage
- This causes the integrator to generate a positive slope ramp
- When this ramp reaches zero, the comparator goes low and stops the counter, whose value represents the analog input



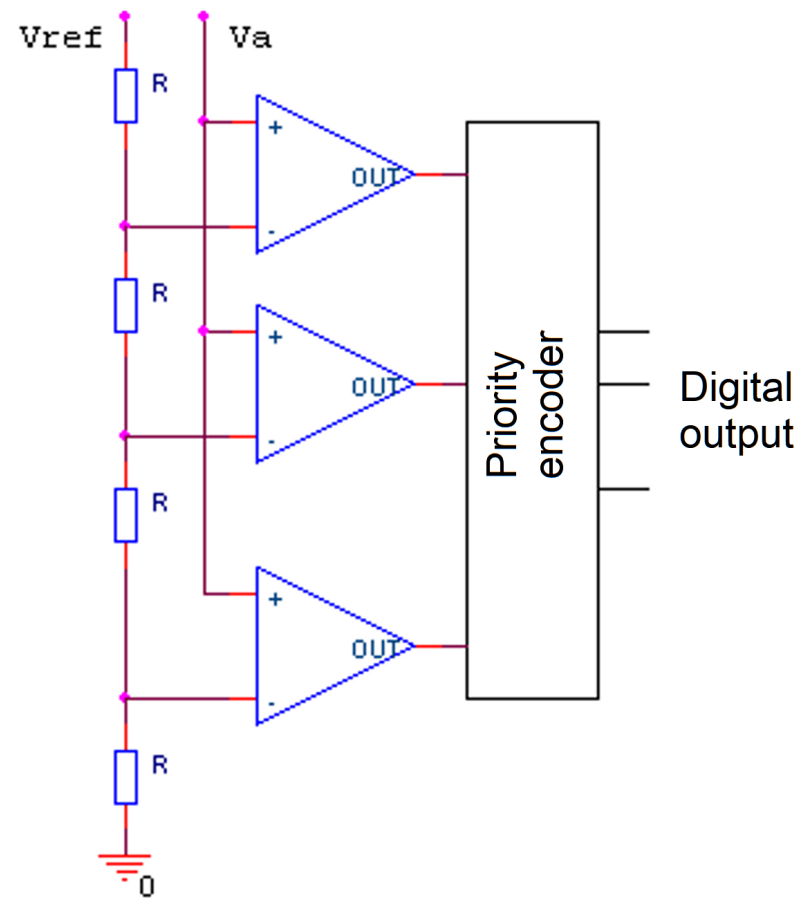


$$T_2 = V_A / V_{ref} * T_1$$



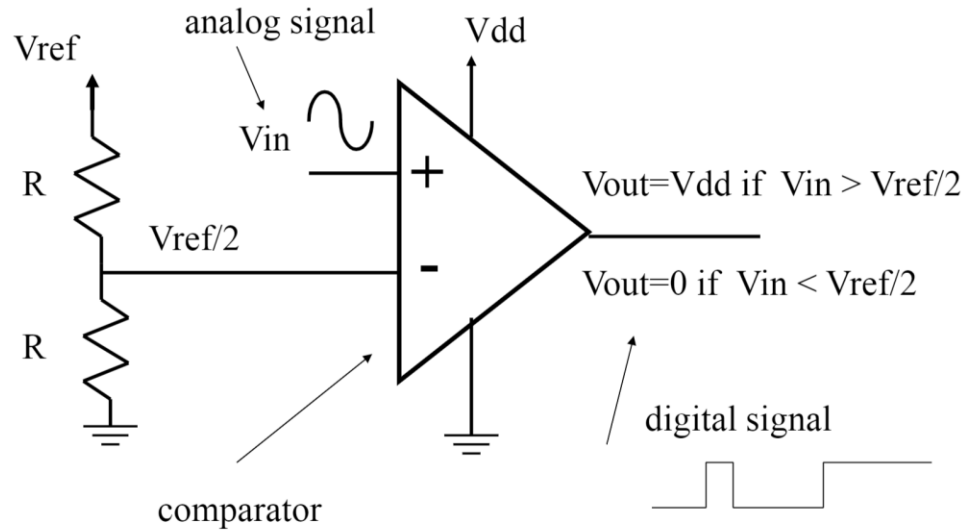
Parallel (flash) ADC

- Basic elements
 - A multiple voltage divider
 - A set of comparators
 - A priority encoder
- Operation
 - Analog input applied to all comparators
 - Priority encoder converts comparator pattern into binary
- E.g.: A 3-bit ADC (see table):
 - For comparator outputs of 0001111, priority encoder generates 100
 - For comparator outputs of 0111111, priority encoder generates 110



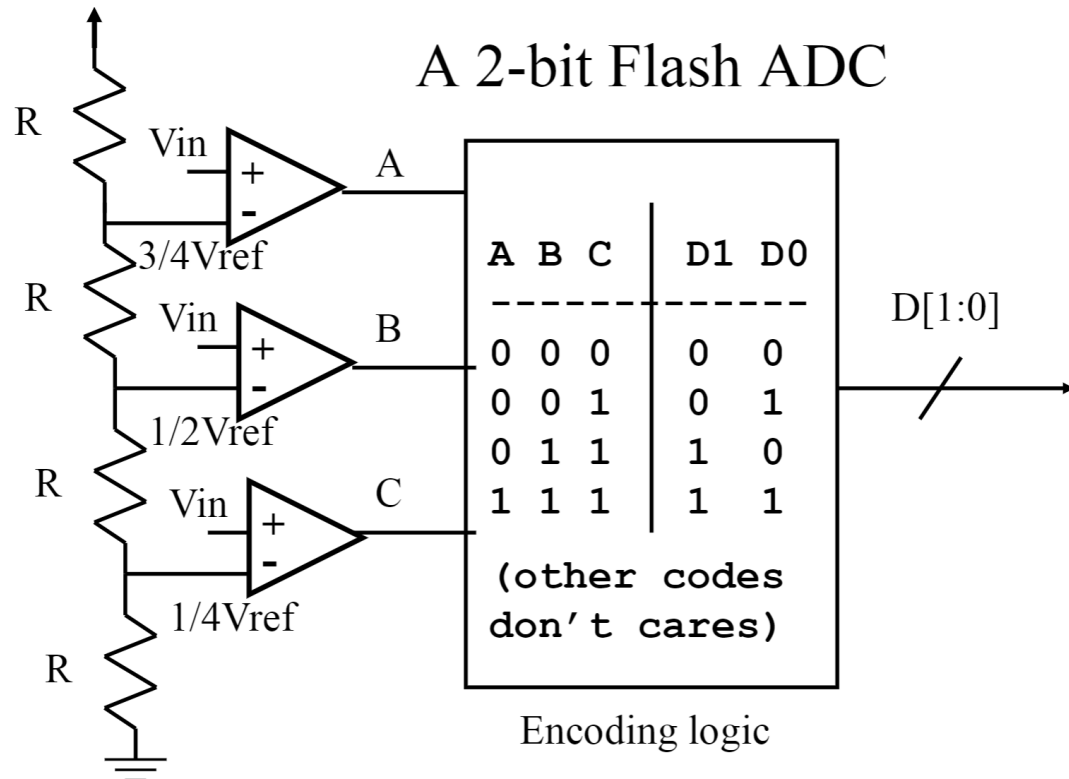
VA	C7	C6	C5	C4	C3	C2	C1	CBA
$VA < 1V$	0	0	0	0	0	0	0	000
$1 < VA < 2V$	0	0	0	0	0	0	1	001
$2 < VA < 3V$	0	0	0	0	0	1	1	010
$3 < VA < 4V$	0	0	0	0	1	1	1	011
$4 < VA < 5V$	0	0	0	1	1	1	1	100
$5 < VA < 6V$	0	0	1	1	1	1	1	101
$6 < VA < 7V$	0	1	1	1	1	1	1	110
$VA > 7V$	1	1	1	1	1	1	1	111

A 1-bit ADC



1-bit and 2-bit ADC examples

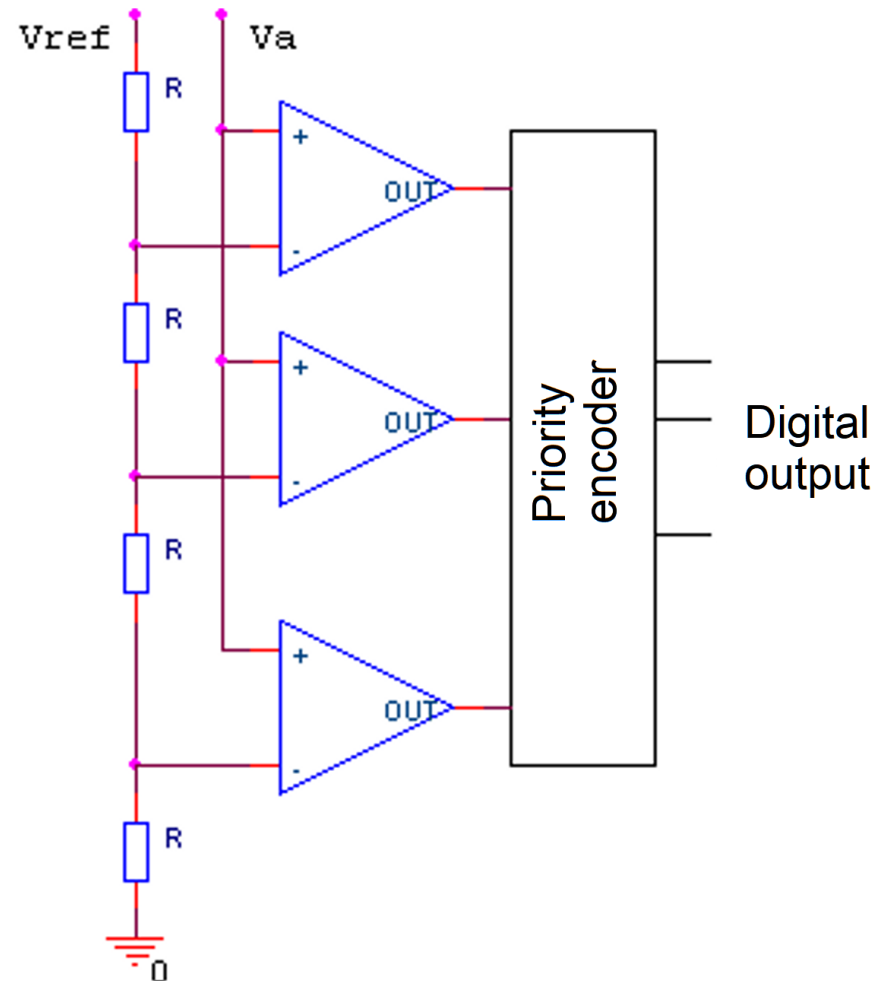
A 2-bit Flash ADC



- Characteristics

- Very fast** (e.g., 8-bit ADCs capable of 20 million conversions/sec)
 - Very expensive for large N** since the number of comparators is $2N-1$

VA	C7	C6	C5	C4	C3	C2	C1	CBA
$VA < 1V$	0	0	0	0	0	0	0	000
$1 < VA < 2V$	0	0	0	0	0	0	1	001
$2 < VA < 3V$	0	0	0	0	0	1	1	010
$3 < VA < 4V$	0	0	0	0	1	1	1	011
$4 < VA < 5V$	0	0	0	1	1	1	1	100
$5 < VA < 6V$	0	0	1	1	1	1	1	101
$6 < VA < 7V$	0	1	1	1	1	1	1	110
$VA > 7V$	1	1	1	1	1	1	1	111



CONVERTERS

Sampling

Signal digitization

Analog-to-Digital Converters

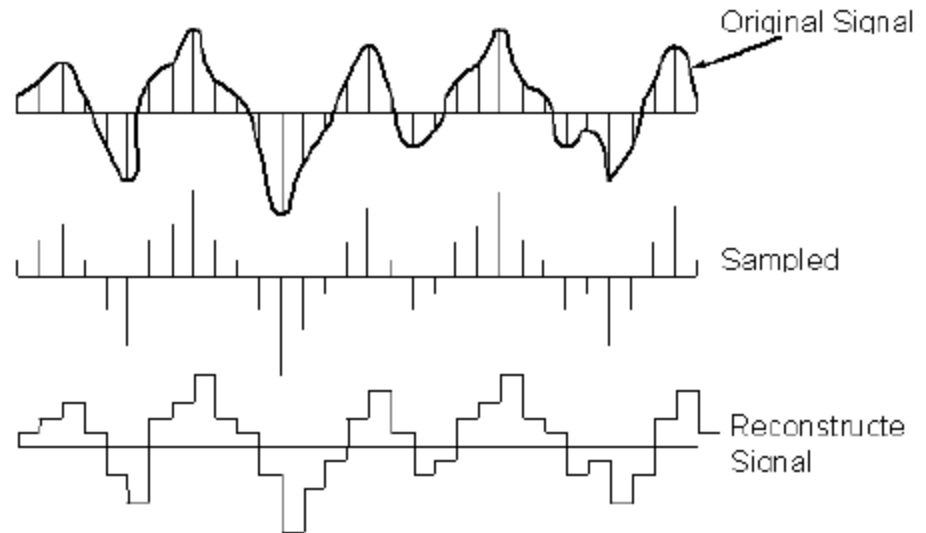
Digital-to-Analog Converters

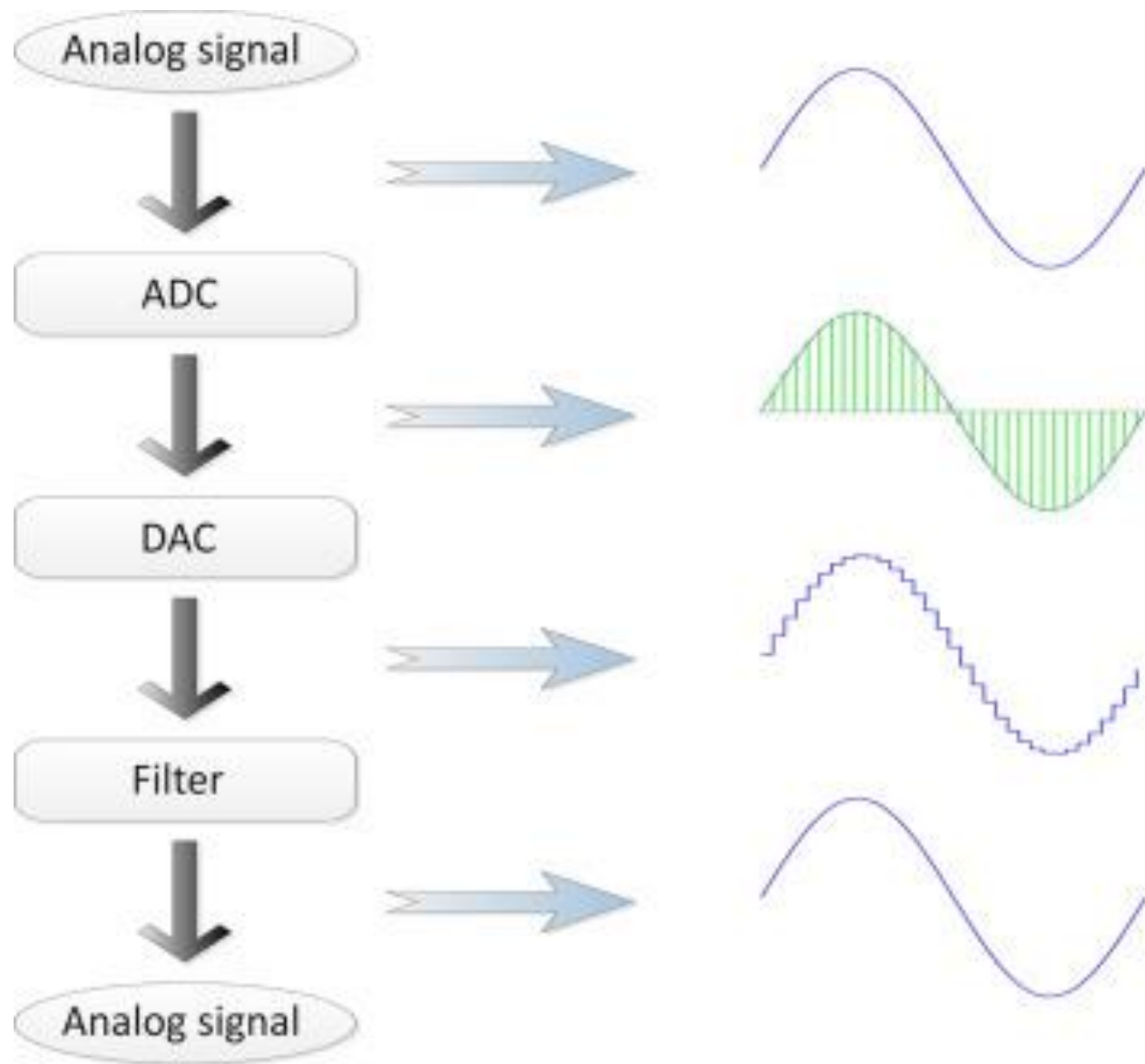
Digital to Analog Converter (DAC)

- A DAC can generate an analog output from a digital input
- **DAC's** performance is **limited** by the **number of samples** it can process and the **number of bits** that is used in converting the digital code into an analog signal
- The output voltage is a sum of voltage components each one twice another

$$V_{out} = U_{ref} \cdot \left(\frac{b_1}{2} + \frac{b_2}{4} + \dots + \frac{b_n}{2^n} \right)$$

- Main types of DACs
 - Binary weighted
 - R-2R ladder
 - Pulse width modulator



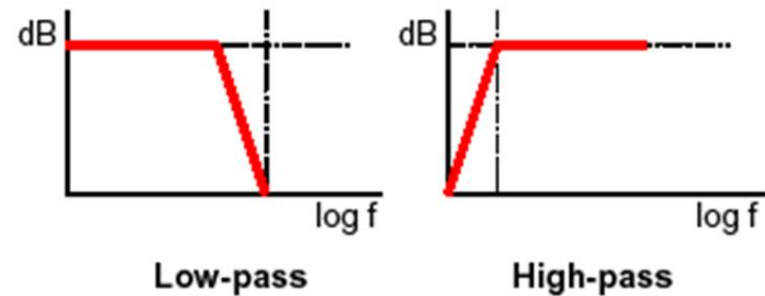


Filters

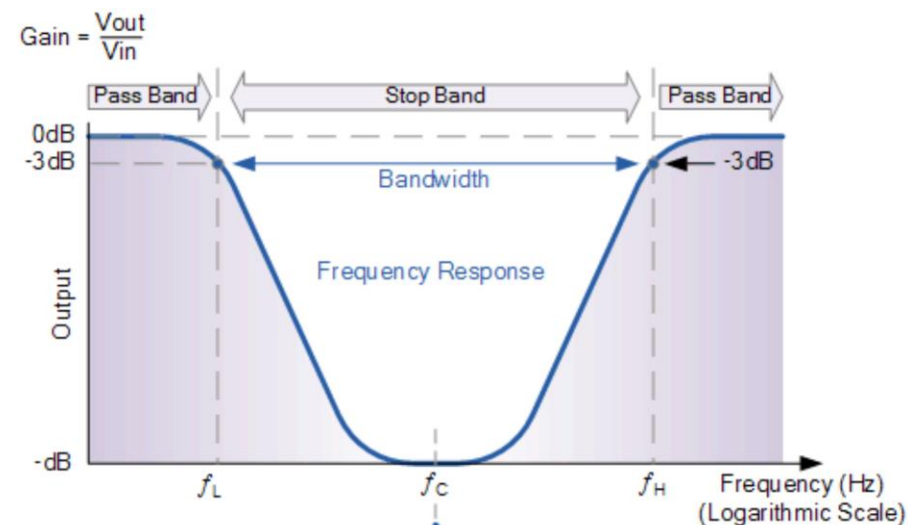
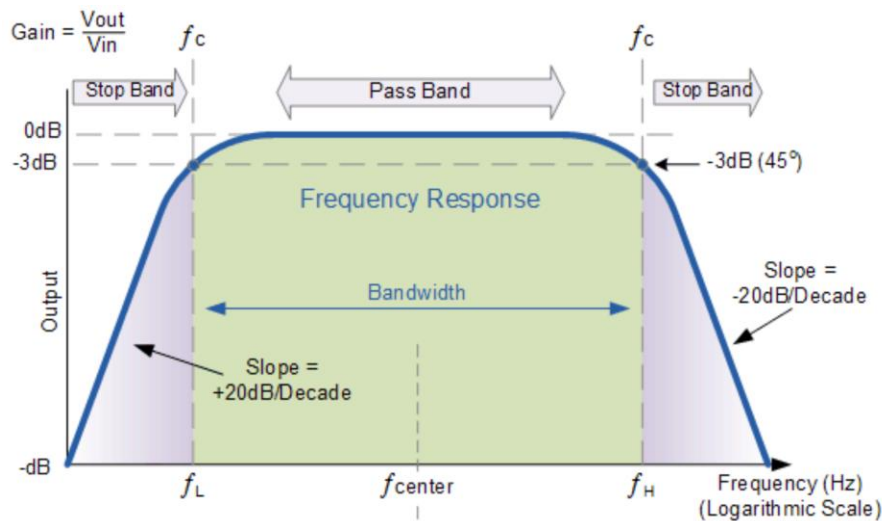
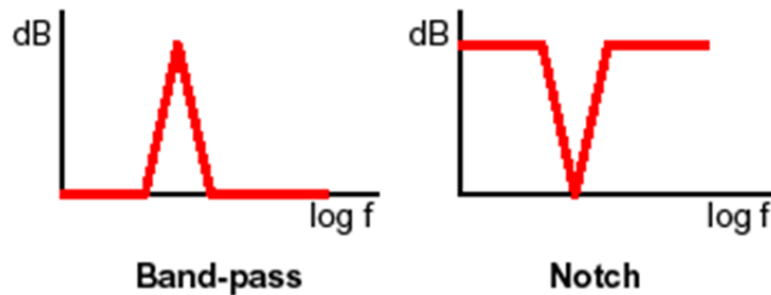
- Filters are used to remove *unwanted* bandwidths from a signal
- Filter classification according to implementation:
 - Active filters include RC networks and op-amps
 - Suitable for low frequency, small signal
 - Active filters are preferred since avoid the bulk and non-linearity of inductors and can have gains greater than 0dB
 - However, active filters require a power supply
 - Passive filters consist of RCL networks
 - Simple, more suitable for frequencies above audio range, where active filters are limited by the op-amp bandwidth
 - Digital filters
 - A digital filter uses a digital processor to perform numerical calculations on sampled values of the signal. The processor may be a general-purpose computer such as a PC, or a specialized DSP (Digital Signal Processor) chip

- Filter classification according to frequency response:

- Low-pass filter
- High-pass filter



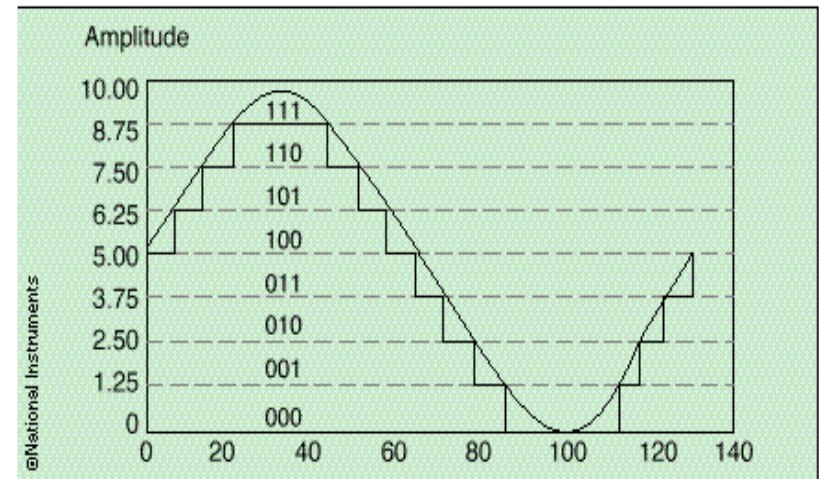
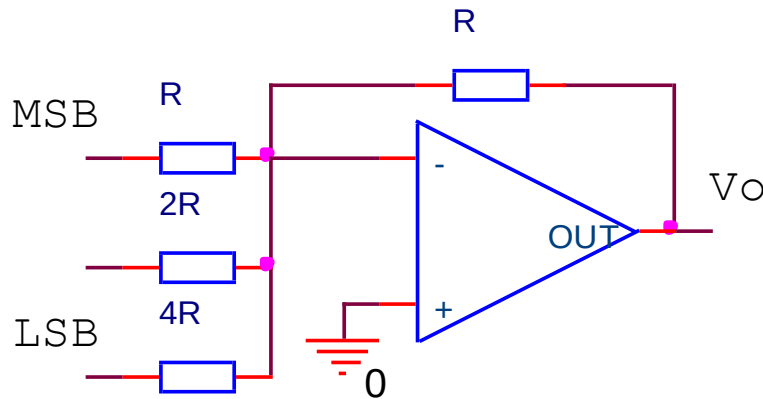
- Band-pass filter
- Band-stop (Notch)



Binary weighted DAC

- Based on the summing op-amp circuit
 - Each input resistor is twice the value of the previous one
 - Inputs are weighted according to their resistors
- Characteristics
 - The lowest value resistor R affects the MSB and must have the highest precision
 - This circuit is impractical for large N since it would require high precision resistors for a wide range

$$V_o = -(V_R + 0.5V_{2R} + 0.25V_{4R} + 0.125V_{8R} + \dots)$$

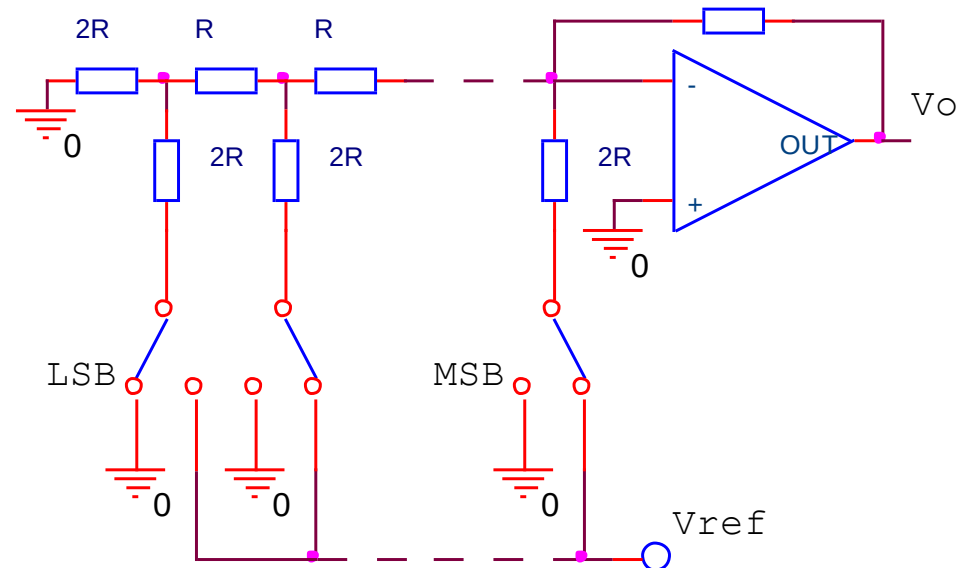


4-bit Binary Weighted D/A Converter Output

Digital Inputs				V _{OUT} Expression	V _{OUT}
D	C	B	A	$1 \cdot V_D + \frac{1}{2} \cdot V_C + \frac{1}{4} \cdot V_B + \frac{1}{8} \cdot V_A$	in Volts
0	0	0	0	$0 \cdot 5 + 0 \cdot 5 + 0 \cdot 5 + 0 \cdot 5$	0
0	0	0	1	$0 \cdot 5 + 0 \cdot 5 + 0 \cdot 5 + \frac{1}{8} \cdot 5$	-0.625
0	0	1	0	$0 \cdot 5 + 0 \cdot 5 + \frac{1}{4} \cdot 5 + 0 \cdot 5$	-1.25
0	0	1	1	$0 \cdot 5 + 0 \cdot 5 + \frac{1}{4} \cdot 5 + \frac{1}{8} \cdot 5$	-1.875
0	1	0	0	$0 \cdot 5 + \frac{1}{2} \cdot 5 + 0 \cdot 5 + 0 \cdot 5$	-2.50
0	1	0	1	$0 \cdot 5 + \frac{1}{2} \cdot 5 + 0 \cdot 5 + \frac{1}{8} \cdot 5$	-3.125
0	1	1	0	$0 \cdot 5 + \frac{1}{2} \cdot 5 + \frac{1}{4} \cdot 5 + 0 \cdot 5$	-3.75
0	1	1	1	$0 \cdot 5 + \frac{1}{2} \cdot 5 + \frac{1}{4} \cdot 5 + \frac{1}{8} \cdot 5$	-4.375
1	0	0	0	$1 \cdot 5 + 0 \cdot 5 + 0 \cdot 5 + 0 \cdot 5$	-5.00
1	0	0	1	$1 \cdot 5 + 0 \cdot 5 + 0 \cdot 5 + \frac{1}{8} \cdot 5$	-5.625
1	0	1	0	$1 \cdot 5 + 0 \cdot 5 + \frac{1}{4} \cdot 5 + 0 \cdot 5$	-6.25
1	0	1	1	$1 \cdot 5 + 0 \cdot 5 + \frac{1}{4} \cdot 5 + \frac{1}{8} \cdot 5$	-6.875
1	1	0	0	$1 \cdot 5 + \frac{1}{2} \cdot 5 + 0 \cdot 5 + 0 \cdot 5$	-7.50
1	1	0	1	$1 \cdot 5 + \frac{1}{2} \cdot 5 + 0 \cdot 5 + \frac{1}{8} \cdot 5$	-8.125
1	1	1	0	$1 \cdot 5 + \frac{1}{2} \cdot 5 + \frac{1}{4} \cdot 5 + 0 \cdot 5$	-8.75
1	1	1	1	$1 \cdot 5 + \frac{1}{2} \cdot 5 + \frac{1}{4} \cdot 5 + \frac{1}{8} \cdot 5$	-9.375

R-2R ladder DAC

- Resistors with similar values, so better for integration
- Operation**
 - When bit k is 1, the corresponding switch is connected to V_{REF}
 - When bit k is 0, the corresponding switch is connected to GND
- Assume all the legs but one are grounded
- The one connected to V_{REF} will generate a current that flows towards the inverting input of the op-amp
- This current is halved by the resistor network at each node
- Therefore, the current contribution of each input is weighted by its position in the binary number



Generalised R-2R DAC Equation

$$V_{OUT} = \frac{V_A + 2V_B + 4V_C + 8V_D + 16V_E + 32V_F + \dots etc}{2^n}$$

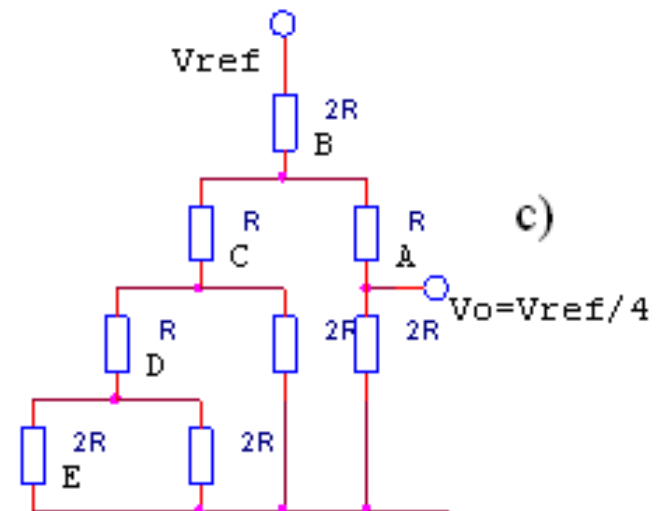
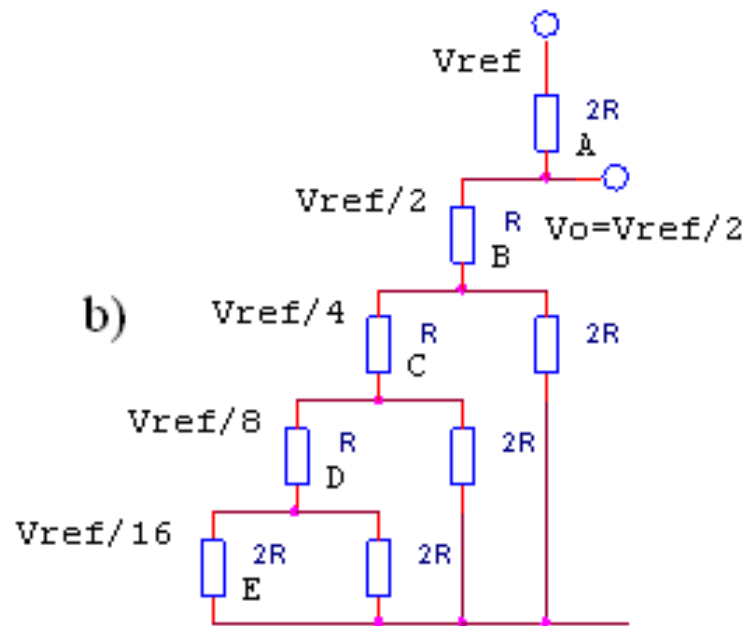
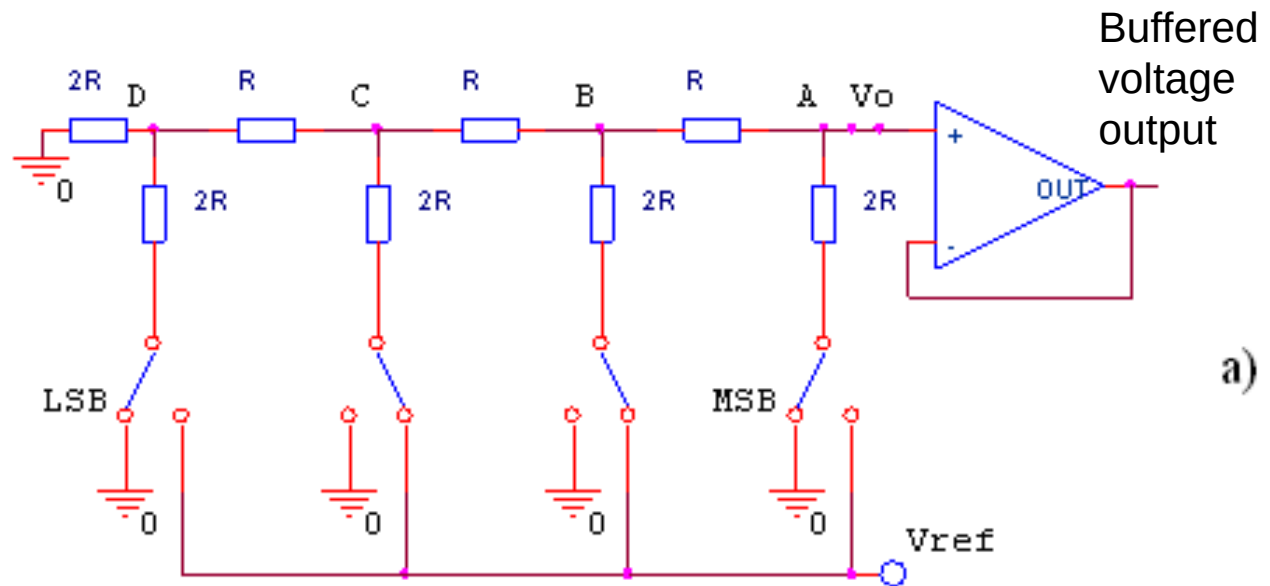
4-bit R-2R D/A Converter Output

n - represents the number of digital inputs

$$V_{LSB} = V_{IN}/2^n$$

Digital Inputs				V _{OUT} Expression	V _{OUT}
D	C	B	A	(8*V _D + 4*V _C + 2*V _B + 1*V _A)/2 ⁴	in Volts
0	0	0	0	(0*5 + 0*5 + 0*5 + 0*5)/16	0
0	0	0	1	(0*5 + 0*5 + 0*5 + 1*5)/16	0.3125
0	0	1	0	(0*5 + 0*5 + 2*5 + 0*5)/16	0.6250
0	0	1	1	(0*5 + 0*5 + 2*5 + 1*5)/16	0.9375
0	1	0	0	(0*5 + 4*5 + 0*5 + 0*5)/16	1.2500
0	1	0	1	(0*5 + 4*5 + 0*5 + 1*5)/16	1.5625
0	1	1	0	(0*5 + 4*5 + 2*5 + 0*5)/16	1.8750
0	1	1	1	(0*5 + 4*5 + 2*5 + 1*5)/16	2.1875
1	0	0	0	(8*5 + 0*5 + 0*5 + 0*5)/16	2.5000
1	0	0	1	(8*5 + 0*5 + 0*5 + 1*5)/16	2.8125
1	0	1	0	(8*5 + 0*5 + 2*5 + 0*5)/16	3.1250
1	0	1	1	(8*5 + 0*5 + 2*5 + 1*5)/16	3.4375
1	1	0	0	(8*5 + 4*5 + 0*5 + 0*5)/16	3.7500
1	1	0	1	(8*5 + 4*5 + 0*5 + 1*5)/16	4.0625
1	1	1	0	(8*5 + 4*5 + 2*5 + 0*5)/16	4.3750
1	1	1	1	(8*5 + 4*5 + 2*5 + 1*5)/16	4.6875

- The R-2R operation is better understood by redrawing the resistor network
- In (b) only the MSB is ON
- In (c) only the next bit to the MSB is ON



Proposed problems

- An analog signal with a maximum frequency of 20kHz is converted to a digital signal. What is the minimum frequency with which this signal should be sampled?

$$f_N = 2 * f_m = 40\text{kHz}$$

- An 8-bit CAN quantifies an analog signal between $V_{\min}=0\text{V}$ and $V_{\max}=10\text{V}$ limits. Calculate the quantization step.